symGCAD — corpus benchmark

All 331 distinct problems of the GCAD corpus: exact generic CAD in Rust on Symbolica, against the original GCAD by Vitaly Magerya

generated 2026-10-12

DRAFT. native GCAD run still in progress (541 of 579 sign cases recorded); Symbolica-vs-FLINT numbers are a snapshot of a running campaign. Regenerate with `benchmark/run_final.sh <symgcad binary>` (see `benchmark/README.md`).

1 Benchmark results

STRICT CONTRACT
303 / 331
independently verified · fixed 124/124 · symbolic 97/124 · core 82/83.

OPT-IN PROJECTIVE
+1
test_gluza_ex33/solve/3 completes only under `projective_sections = true` (§1.2); 2 of 2 opt-in rows verified. **Not** counted in the strict total.

NATIVE GCAD, 300 S/CASE
271 / 331
groups completed · 30 timeout · 2 out of memory (8 GiB) · 28 not run yet cases 479/579 complete.

GEOMETRIC MEAN RATIO
0.059
symGCAD solve / native time on the 271 groups both complete (17× faster); 267 of 271 faster, 2 more than 2× slower.

ONLY SYMGCAD COMPLETES
14
groups verified by symGCAD where native GCAD timed out or ran out of memory. Summed times on the common set: 176.5 s vs 752.6 s.

What is compared. Each row is one distinct original problem (aliases merged; split polynomials kept). **symGCAD** solves the full problem in one fresh process with its recipe (engine, ordering policy, options from the 2026-10-10 audit, `benchmark/campaign/symgcad_inputs/`), then `symgcad verify` re-checks the exported proof in a second fresh process. Only a verified `complete_generic` result counts. **Native** is the unmodified GCAD (`fb6ebe3`, FLINT 3.6) through the unmodified bridge `reference/adapter.py`: each sign case of the split (`F > 0` and `F < 0` added as constraints, as in the earlier parity campaign) is solved in its own process, capped at 300 s and 8 GiB; the native time is the sum over sign cases and includes Python start-up and SymPy parsing. The **ratio** is symGCAD solve / native time; green < 1 (symGCAD faster), yellow 1–2, red > 2. Most corpus problems are tiny, and there the native time is dominated by Python start-up. On the 39 groups where native GCAD needs at least 1 s of internal time, the geometric mean ratio is 0.069 against native process time and 0.103 against native internal time (9.8× faster without any start-up cost on the native side).

Columns. Order is the variable order of the exported result (projection eliminates from the right); **H** marks the homogeneous anchor fixed to 1 before solving. **Method** is the engine and ordering policy (`mods` = GCAD's heuristic, `given` = recipe order, `mcts` = bounded search inside the timed solve). **Cells** are verified raw cells (no merging; GCAD merges, so cell counts are not comparable); † marks the 53 groups whose count differs from the 2026-10-10 audit (both verified; recipes and code changed since). Times are wall seconds on one pinned CPU of an AMD EPYC 9754, with cold caches (fresh checkpoint directory and `TMPDIR` per run); verified rows are sorted by solve time, then the incomplete ones. **Provenance:** symGCAD binary `4f212724f4a9` (303 groups) run 2026-10-12T12:03–12:09 UTC native launch 2026-10-12T10:27 UTC, upstream `fb6ebe3`. Every row's binary hash, input hash, date and record path are in `report-data.json`.

1.1 All 331 problems

#	Problem	Scope	Status	Method · ordering	Order (H = anchor)	Cells	Solve s	Verify s	Native s	Ratio
1	finiteAx12_6_190_7_0__0_2_1_1_1_1_0_1_0/fixed	fixed	verified	monotone · mods	x4 x7 x5 x3 x2 x1 · H=x4	1	0.0344	0.0292	1.62	0.021
2	finiteB_4_277_5_0__2_0_1_0_1_0_0_0_1/fixed	fixed	verified	monotone · mods	x4 x2 x0 x8 · H=x4	4†	0.0431	0.0316	1.22	0.035
3	finiteE_4_15__4_0__1_1_1_1/fixed	fixed	verified	monotone · mods	x0 x2 x3 x1 · H=x0	5	0.0436	0.0281	1.23	0.036
4	finiteE_2_10__3_0__0_2_0_1/fixed	fixed	verified	monotone · mods	x1 x3 · H=x1	1	0.0436	0.0280	1.22	0.036
5	finiteAx12_5_376_6_0__0_0_0_2_1_1_1_0_1/fixed	fixed	verified	monotone · mods	x4 x6 x8 x5 x3 · H=x4	5†	0.0439	0.0300	1.41	0.031
6	finiteAx12_5_316_10_0__0_0_6_1_1_1_0_0_1/fixed	fixed	verified	monotone · mods	x4 x2 x8 x5 x3 · H=x4	6	0.0445	0.0292	1.43	0.031
7	finiteA_5_301_6_0__2_0_1_1_0_1_0_0_1/fixed	fixed	verified	monotone · mods	x0 x2 x8 x3 x5 · H=x0	5†	0.0446	0.0302	1.42	0.031
8	finiteA_6_190_7_0__0_2_1_1_1_1_0_1_0/fixed	fixed	verified	monotone · mods	x4 x7 x5 x3 x2 x1 · H=x4	1	0.0447	0.0290	1.82	0.025
9	finiteAx12_5_410_10_0__0_6_0_1_1_0_0_1_1/fixed	fixed	verified	monotone · mods	x4 x7 x8 x3 x1 · H=x4	1	0.0449	0.0279	1.22	0.037
10	finiteAx12_4_297_5_0__2_0_0_1_0_1_0_0_1/fixed	fixed	verified	monotone · mods	x0 x3 x8 x5 · H=x0	1	0.0449	0.0292	1.62	0.028
11	finiteA_6_318_7_0__0_2_1_1_1_1_0_0_1/fixed	fixed	verified	monotone · mods	x4 x2 x8 x3 x1 x5 · H=x4	7	0.0450	0.0295	2.42	0.019
12	finiteA_5_87_6_0__2_1_1_0_1_0_1_0_0/fixed	fixed	verified	monotone · mods	x4 x6 x2 x0 x1 · H=x4	19†	0.0453	0.0299	1.22	0.037
13	finiteA_5_171_6_0__2_1_0_1_0_1_0_1_0/fixed	fixed	verified	monotone · mods	x0 x3 x1 x7 x5 · H=x0	1	0.0454	0.0290	1.22	0.037
14	finiteA_4_89_5_0__2_0_0_1_1_0_1_0_0/fixed	fixed	verified	monotone · mods	x4 x6 x3 x0 · H=x4	5†	0.0455	0.0291	1.82	0.025
15	finiteAx12_5_31_6_0__2_1_1_1_1_0_0_0_0/fixed	fixed	verified	monotone · mods	x4 x2 x0 x3 x1 · H=x4	7	0.0456	0.0303	2.02	0.023
16	finiteE_2_9__3_0__2_0_0_1/fixed	fixed	verified	monotone · mods	x0 x3 · H=x0	1	0.0456	0.0280	1.22	0.038
17	finiteA_4_344_5_0__0_0_0_2_1_0_1_0_1/fixed	fixed	verified	monotone · mods	x4 x6 x8 x3 · H=x4	4†	0.0457	0.0309	1.22	0.038
18	finiteA_3_276_4_0__0_0_2_0_1_0_0_0_1/symbolic	symbolic	verified	monotone · mods	s x4 x2 x8 · H=x4	4	0.0459	0.0287	1.23	0.037
19	finiteA_4_312_5_0__0_0_0_2_1_1_0_0_1/symbolic	symbolic	verified	monotone · mods	s t x4 x5 x3 x8 · H=x4	9	0.0459	0.0298	1.62	0.028
20	finiteA_4_29_5_0__2_0_1_1_1_0_0_0_0/fixed	fixed	verified	monotone · mods	x4 x2 x0 x3 · H=x4	6	0.0461	0.0293	1.42	0.033

#	Problem	Scope	Status	Method · ordering	Order (H = anchor)	Cells	Solve s	Verify s	Native s	Ratio
21	finiteA_5_412_6_0__0_0_2_1_1_0_0_1_1/fixed	fixed	verified	monotone · mods	x4 x2 x8 x7 x3 · H=x4	6	0.0462	0.0311	1.22	0.038
22	finiteA_4_154_5_0__0_2_0_1_1_0_0_1_0/symbolic	symbolic	verified	monotone · mods	s t x4 x3 x1 x7 · H=x4	9†	0.0462	0.0284	1.23	0.038
23	finiteE_3_13__3_0__1_0_1_1/fixed	fixed	verified	monotone · mods	x0 x2 x3 · H=x0	4	0.0462	0.0310	1.21	0.038
24	finiteE_2_5__3_0__2_0_1_0/fixed	fixed	verified	monotone · mods	x0 x2 · H=x0	3	0.0463	0.0279	1.62	0.029
25	finiteA_5_313_6_0__2_0_0_1_1_1_0_0_1/fixed	fixed	verified	monotone · mods	x4 x5 x8 x3 x0 · H=x4	1	0.0463	0.0294	1.22	0.038
26	finiteE_3_7__3_0__1_1_1_0/fixed	fixed	verified	monotone · mods	x0 x2 x1 · H=x0	4	0.0464	0.0282	1.22	0.038
27	finiteA_4_293_5_0__2_0_1_0_0_1_0_0_1/fixed	fixed	verified	monotone · mods	x0 x2 x8 x5 · H=x0	4†	0.0464	0.0293	1.41	0.033
28	finiteAx12_4_282_5_0__0_2_0_1_1_0_0_0_1/fixed	fixed	verified	monotone · mods	x4 x8 x3 x1 · H=x4	1	0.0465	0.0293	1.22	0.038
29	finiteAx12_5_313_6_0__2_0_0_1_1_1_0_0_1/fixed	fixed	verified	monotone · mods	x4 x5 x8 x3 x0 · H=x4	1	0.0466	0.0297	1.62	0.029
30	finiteAx12_4_312_5_0__0_0_0_2_1_1_0_0_1/fixed	fixed	verified	monotone · mods	x4 x5 x8 x3 · H=x4	1	0.0467	0.0290	1.63	0.029
31	finiteA_5_155_6_0__2_1_0_1_1_0_0_1_0/fixed	fixed	verified	monotone · mods	x4 x3 x1 x0 x7 · H=x4	1	0.0467	0.0305	1.81	0.026
32	finiteAx12_5_155_6_0__2_1_0_1_1_0_0_1_0/fixed	fixed	verified	monotone · mods	x4 x3 x1 x0 x7 · H=x4	1	0.0467	0.0300	1.22	0.038
33	finiteAx12_5_171_6_0__2_1_0_1_0_1_0_1_0/fixed	fixed	verified	monotone · mods	x0 x3 x1 x7 x5 · H=x0	1	0.0468	0.0293	1.61	0.029
34	finiteAx12_5_314_6_0__0_2_0_1_1_1_0_0_1/fixed	fixed	verified	monotone · mods	x4 x5 x8 x3 x1 · H=x4	1	0.0468	0.0274	1.21	0.039
35	finiteA_4_284_5_0__0_0_2_1_1_0_0_0_1/fixed	fixed	verified	monotone · mods	x4 x2 x8 x3 · H=x4	5	0.0469	0.0302	1.61	0.029
36	finiteA_5_376_6_0__0_0_0_2_1_1_1_0_1/fixed	fixed	verified	monotone · mods	x4 x6 x8 x5 x3 · H=x4	5†	0.0469	0.0306	1.21	0.039
37	finiteAx12_5_286_6_0__0_2_1_1_1_0_0_0_1/fixed	fixed	verified	monotone · mods	x4 x2 x8 x3 x1 · H=x4	6	0.0470	0.0294	1.82	0.026
38	finiteA_5_286_6_0__0_2_1_1_1_0_0_0_1/fixed	fixed	verified	monotone · mods	x4 x2 x8 x3 x1 · H=x4	6	0.0471	0.0299	1.42	0.033
39	finiteA_3_25_4_0__2_0_0_1_1_0_0_0_0/symbolic	symbolic	verified	monotone · mods	s x4 x3 x0 · H=x4	1	0.0472	0.0281	1.23	0.038
40	finiteCx12_7_247_10_0__4_1_1_0_1_1_1_1_0/fixed	fixed	verified	monotone · mods	x4 x6 x5 x7 x2 x1 x0 · H=x4	1	0.0473	0.0289	timeout	—
41	finiteAx12_5_301_6_0__2_0_1_1_0_1_0_0_1/fixed	fixed	verified	monotone · mods	x0 x2 x8 x3 x5 · H=x0	5†	0.0473	0.0288	1.21	0.039
42	finiteE_3_11__3_0__1_1_0_1/fixed	fixed	verified	monotone · mods	x0 x3 x1 · H=x0	1	0.0474	0.0283	1.83	0.026
43	finiteAx12_3_296_4_0__0_0_0_2_0_1_0_0_1/fixed	fixed	verified	monotone · mods	x3 x8 x5 · H=x3	1	0.0474	0.0290	1.62	0.029
44	finiteA_4_284_5_0__0_0_2_1_1_0_0_0_1/symbolic	symbolic	verified	monotone · mods	s x4 x2 x8 x3 · H=x4	5	0.0474	0.0459	1.23	0.039
45	finiteAx12_6_414_10_0__0_5_1_1_1_0_0_1_1/fixed	fixed	verified	monotone · mods	x4 x2 x8 x3 x1 x7 · H=x4	7	0.0474	0.0280	5.03	0.0094
46	finiteA_3_21_4_0__2_0_1_0_1_0_0_0_0/symbolic	symbolic	verified	monotone · mods	s x4 x2 x0 · H=x4	5	0.0475	0.0281	1.43	0.033
47	finiteA_4_85_5_0__2_0_1_0_1_0_1_0_0/fixed	fixed	verified	monotone · mods	x4 x6 x2 x0 · H=x4	15	0.0475	0.0302	1.42	0.034
48	finiteA_5_316_10_0__0_0_6_1_1_1_0_0_1/fixed	fixed	verified	monotone · mods	x4 x2 x8 x5 x3 · H=x4	6	0.0476	0.0293	1.42	0.034
49	finiteAx12_4_154_5_0__0_2_0_1_1_0_0_1_0/symbolic	symbolic	verified	monotone · mods	t s x4 x3 x1 x7 · H=x4	9†	0.0476	0.0299	1.43	0.033
50	finiteA_5_410_10_0__0_6_0_1_1_0_0_1_1/fixed	fixed	verified	monotone · mods	x4 x7 x8 x3 x1 · H=x4	1	0.0476	0.0292	1.42	0.034
51	finiteA_3_26_4_0__0_2_0_1_1_0_0_0_0/symbolic	symbolic	verified	monotone · mods	s t x4 x3 x1 · H=x4	8	0.0477	0.0286	1.22	0.039
52	finiteA_4_170_5_0__0_2_0_1_0_1_0_1_0/fixed	fixed	verified	monotone · mods	x1 x7 x5 x3 · H=x1	1	0.0478	0.0301	1.42	0.034
53	finiteA_4_282_5_0__0_2_0_1_1_0_0_0_1/fixed	fixed	verified	monotone · mods	x4 x8 x3 x1 · H=x4	1	0.0479	0.0293	1.42	0.034
54	finiteA_4_154_5_0__0_2_0_1_1_0_0_1_0/fixed	fixed	verified	monotone · mods	x4 x7 x3 x1 · H=x4	1	0.0480	0.0300	1.42	0.034
55	finiteA_4_312_5_0__0_0_0_2_1_1_0_0_1/fixed	fixed	verified	monotone · mods	x4 x5 x8 x3 · H=x4	1	0.0480	0.0296	1.22	0.039
56	finiteA_5_314_6_0__0_2_0_1_1_1_0_0_1/fixed	fixed	verified	monotone · mods	x4 x5 x8 x3 x1 · H=x4	1	0.0482	0.0306	1.82	0.026
57	finiteA_4_297_5_0__2_0_0_1_0_1_0_0_1/fixed	fixed	verified	monotone · mods	x0 x3 x8 x5 · H=x0	1	0.0483	0.0285	1.22	0.040
58	finiteA_2_17_3_0__2_0_0_0_1_0_0_0_0/symbolic	symbolic	verified	monotone · mods	s x4 x0 · H=x4	1	0.0483	0.0280	1.22	0.039
59	finiteA_4_27_5_0__2_1_0_1_1_0_0_0_0/fixed	fixed	verified	monotone · mods	x4 x3 x1 x0 · H=x4	1	0.0483	0.0290	1.42	0.034
60	finiteB_5_213_6_0__2_0_1_0_1_0_1_1_0/fixed	fixed	verified	monotone · mods	x4 x7 x6 x2 x0 · H=x4	10	0.0483	0.0280	1.41	0.034
61	finiteF_4_15__4_0__1_1_1_1/fixed	fixed	verified	monotone · mods	x0 x3 x2 x1 · H=x0	1	0.0483	0.0280	1.23	0.039
62	finiteB_5_339_6_0__2_1_0_0_1_0_1_0_1/fixed	fixed	verified	monotone · mods	x4 x8 x6 x0 x1 · H=x4	6	0.0483	0.0282	1.42	0.034
63	finiteB_3_21_4_0__2_0_1_0_1_0_0_0_0/fixed	fixed	verified	monotone · mods	x4 x2 x0 · H=x4	3	0.0483	0.0298	1.62	0.030
64	finiteA_5_157_6_0__2_0_1_1_1_0_0_1_0/fixed	fixed	verified	monotone · mods	x4 x2 x0 x3 x7 · H=x4	7†	0.0483	0.0302	1.22	0.040
65	finiteB_5_341_6_0__2_0_1_0_1_0_1_0_1/fixed	fixed	verified	monotone · mods	x4 x8 x6 x2 x0 · H=x4	11	0.0484	0.0287	1.42	0.034
66	finiteA_4_153_5_0__2_0_0_1_1_0_0_1_0/symbolic	symbolic	verified	monotone · mods	s x4 x7 x3 x0 · H=x4	1	0.0486	0.0284	1.23	0.040
67	finiteAx12_6_318_7_0__0_2_1_1_1_1_0_0_1/fixed	fixed	verified	monotone · mods	x4 x2 x8 x3 x1 x5 · H=x4	7	0.0486	0.0304	2.62	0.019
68	finiteA_5_345_6_0__2_0_0_1_1_0_1_0_1/fixed	fixed	verified	monotone · mods	x4 x6 x0 x8 x3 · H=x4	12†	0.0487	0.0290	1.82	0.027
69	finiteA_3_26_4_0__0_2_0_1_1_0_0_0_0/fixed	fixed	verified	monotone · mods	x4 x3 x1 · H=x4	1	0.0488	0.0287	1.22	0.040
70	finiteC_6_95_7_0__2_1_1_1_1_0_1_0_0/fixed	fixed	verified	monotone · mods	x4 x3 x2 x1 x0 x6 · H=x4	1	0.0488	0.0292	1.62	0.030
71	finiteB_5_279_6_0__2_1_1_0_1_0_0_0_1/fixed	fixed	verified	monotone · mods	x4 x2 x0 x1 x8 · H=x4	5†	0.0489	0.0282	1.41	0.035
72	finiteC_6_215_11_0__6_1_1_0_1_0_1_1_0/fixed	fixed	verified	monotone · mods	x4 x2 x1 x0 x7 x6 · H=x4	1	0.0489	0.0293	5.63	0.0087
73	finiteA_4_344_5_0__0_0_0_2_1_0_1_0_1/symbolic	symbolic	verified	monotone · mods	s x4 x6 x8 x3 · H=x4	4†	0.0490	0.0294	1.23	0.040
74	finiteAx12_4_27_5_0__2_1_0_1_1_0_0_0_0/fixed	fixed	verified	monotone · mods	x4 x3 x1 x0 · H=x4	1	0.0490	0.0293	1.22	0.040
75	finiteC_6_231_7_0__2_1_1_0_0_1_1_1_0/fixed	fixed	verified	monotone · mods	x0 x2 x1 x7 x6 x5 · H=x0	1	0.0491	0.0300	3.47	0.014
76	finiteA_5_310_6_0__0_2_1_0_1_1_0_0_1/fixed	fixed	verified	monotone · mods	x4 x2 x8 x5 x1 · H=x4	6	0.0491	0.0281	1.62	0.030
77	finiteAx12_4_170_5_0__0_2_0_1_0_1_0_1_0/fixed	fixed	verified	monotone · mods	x1 x7 x5 x3 · H=x1	1	0.0492	0.0287	1.62	0.030

#	Problem	Scope	Status	Method · ordering	Order (H = anchor)	Cells	Solve s	Verify s	Native s	Ratio
78	finiteC_5_227_6_0__2_1_0_0_0_1_1_1_0/fixed	fixed	verified	monotone · mods	x0 x7 x6 x5 x1 · H=x0	1	0.0493	0.0294	1.42	0.035
79	finiteA_3_296_4_0__0_0_0_2_0_1_0_0_1/symbolic	symbolic	verified	monotone · mods	s t x3 x5 x8 · H=x3	6	0.0494	0.0287	1.22	0.040
80	finiteAx12_4_27_5_0__2_1_0_1_1_0_0_0_0/symbolic	symbolic	verified	monotone · mods	t s x4 x3 x1 x0 · H=x4	9	0.0495	0.0297	1.22	0.040
81	finiteAx12_4_154_5_0__0_2_0_1_1_0_0_1_0/fixed	fixed	verified	monotone · mods	x4 x7 x3 x1 · H=x4	1	0.0497	0.0295	1.63	0.031
82	finiteA_3_280_4_0__0_0_0_2_1_0_0_0_1/symbolic	symbolic	verified	monotone · mods	s x4 x8 x3 · H=x4	1	0.0498	0.0313	1.22	0.041
83	finiteBx12_5_279_6_0__2_1_1_0_1_0_0_0_1/fixed	fixed	verified	monotone · mods	x4 x2 x0 x1 x8 · H=x4	5†	0.0500	0.0295	1.22	0.041
84	finiteA_4_153_5_0__2_0_0_1_1_0_0_1_0/fixed	fixed	verified	monotone · mods	x4 x7 x3 x0 · H=x4	1	0.0501	0.0306	1.62	0.031
85	finiteA_4_89_5_0__2_0_0_1_1_0_1_0_0/symbolic	symbolic	verified	monotone · mods	s x4 x6 x3 x0 · H=x4	5†	0.0502	0.0282	1.63	0.031
86	finiteCx12_6_438_7_0__0_2_1_0_1_1_0_1_1/fixed	fixed	verified	monotone · mods	x4 x8 x7 x5 x2 x1 · H=x4	1	0.0502	0.0302	1.61	0.031
87	finiteA_4_27_5_0__2_1_0_1_1_0_0_0_0/symbolic	symbolic	verified	monotone · mods	s t x4 x3 x1 x0 · H=x4	9	0.0503	0.0295	1.22	0.041
88	finiteC_6_438_7_0__0_2_1_0_1_1_0_1_1/fixed	fixed	verified	monotone · mods	x4 x8 x7 x5 x2 x1 · H=x4	1	0.0504	0.0306	1.62	0.031
89	finiteC_5_158_6_0__0_2_1_1_1_0_0_1_0/fixed	fixed	verified	monotone · mods	x4 x3 x2 x1 x7 · H=x4	1	0.0504	0.0317	1.41	0.036
90	finiteC_7_247_10_0__4_1_1_0_1_1_1_1_0/fixed	fixed	verified	monotone · mods	x4 x6 x5 x7 x2 x1 x0 · H=x4	1	0.0505	0.0291	timeout	—
91	finiteC_5_412_6_0__0_0_2_1_1_0_0_1_1/fixed	fixed	verified	monotone · mods	x4 x8 x7 x3 x2 · H=x4	1	0.0509	0.0297	1.42	0.036
92	finiteB_4_85_5_0__2_0_1_0_1_0_1_0_0/fixed	fixed	verified	monotone · mods	x4 x2 x0 x6 · H=x4	4†	0.0509	0.0280	1.42	0.036
93	finiteA_5_121_6_0__2_0_0_1_1_1_1_0_0/fixed	fixed	verified	monotone · mods	x4 x6 x3 x5 x0 · H=x4	6†	0.0509	0.0291	2.03	0.025
94	finiteBx12_5_339_6_0__2_1_0_0_1_0_1_0_1/fixed	fixed	verified	monotone · mods	x4 x8 x6 x0 x1 · H=x4	6	0.0509	0.0293	1.42	0.036
95	finiteB_4_197_5_0__2_0_1_0_0_0_1_1_0/fixed	fixed	verified	monotone · mods	x0 x7 x6 x2 · H=x0	13	0.0509	0.0281	1.22	0.042
96	finiteCx12_6_231_7_0__2_1_1_0_0_1_1_1_0/fixed	fixed	verified	monotone · mods	x0 x2 x1 x7 x6 x5 · H=x0	1	0.0509	0.0280	3.02	0.017
97	finiteAx12_5_155_6_0__2_1_0_1_1_0_0_1_0/symbolic	symbolic	verified	monotone · mods	t s x4 x3 x1 x0 x7 · H=x4	10†	0.0512	0.0286	1.42	0.036
98	finiteAx12_3_26_4_0__0_2_0_1_1_0_0_0_0/fixed	fixed	verified	monotone · mods	x4 x3 x1 · H=x4	1	0.0512	0.0293	1.42	0.036
99	finiteB_4_277_5_0__2_0_1_0_1_0_0_0_1/symbolic	symbolic	verified	monotone · mods	s x4 x2 x0 x8 · H=x4	4†	0.0514	0.0290	1.42	0.036
100	finiteA_3_25_4_0__2_0_0_1_1_0_0_0_0/fixed	fixed	verified	monotone · mods	x4 x3 x0 · H=x4	1	0.0514	0.0281	1.22	0.042
101	test_greedy_sotd_order/solve/2	core	verified	optimized · mods	x y	7	0.0516	0.0309	0.614	0.084
102	test_y2_minus_x/solve/1	core	verified	optimized · mods	y x	1	0.0518	0.0309	0.610	0.085
103	finiteA_3_296_4_0__0_0_0_2_0_1_0_0_1/fixed	fixed	verified	monotone · mods	x3 x8 x5 · H=x3	1	0.0518	0.0306	1.21	0.043
104	finiteA_3_280_4_0__0_0_0_2_1_0_0_0_1/fixed	fixed	verified	monotone · mods	x4 x8 x3 · H=x4	1	0.0519	0.0307	1.41	0.037
105	finiteCx12_5_158_6_0__0_2_1_1_1_0_0_1_0/fixed	fixed	verified	monotone · mods	x4 x3 x2 x1 x7 · H=x4	1	0.0519	0.0281	1.21	0.043
106	finiteA_5_31_6_0__2_1_1_1_1_0_0_0_0/fixed	fixed	verified	monotone · mods	x4 x2 x0 x3 x1 · H=x4	7	0.0520	0.0298	1.22	0.043
107	finiteE_3_7__3_0__1_1_1_0/symbolic	symbolic	verified	monotone · mods	s x0 x2 x1 · H=x0	4	0.0522	0.0310	1.22	0.043
108	finiteA_4_278_5_0__0_2_1_0_1_0_0_0_1/fixed	fixed	verified	monotone · mods	x4 x2 x8 x1 · H=x4	5	0.0524	0.0303	1.22	0.043
109	finiteA_4_23_5_0__2_1_1_0_1_0_0_0_0/fixed	fixed	verified	monotone · mods	x4 x2 x0 x1 · H=x4	6	0.0525	0.0302	1.62	0.032
110	finiteC_5_31_6_0__2_1_1_1_1_0_0_0_0/fixed	fixed	verified	monotone · mods	x4 x3 x2 x1 x0 · H=x4	1	0.0527	0.0299	1.41	0.037
111	finiteA_4_293_5_0__2_0_1_0_0_1_0_0_1/symbolic	symbolic	verified	monotone · mods	s x0 x2 x8 x5 · H=x0	4†	0.0528	0.0298	1.23	0.043
112	test_ex412_box/solve/1	core	verified	optimized · mods	x4 x1 p1s s13 x3 x2 s12	2	0.0528	0.0307	0.610	0.087
113	finiteA_4_23_5_0__2_1_1_0_1_0_0_0_0/symbolic	symbolic	verified	monotone · mods	s x4 x2 x0 x1 · H=x4	6	0.0531	0.0297	1.23	0.043
114	finiteA_3_276_4_0__0_0_2_0_1_0_0_0_1/fixed	fixed	verified	monotone · mods	x4 x2 x8 · H=x4	4	0.0531	0.0294	1.21	0.044
115	test_triangle2l_split/order/1	core	verified	optimized · mods	x4 x3 x5 x2 x1 x0 · H=x4	4	0.0533	0.0297	*not run*	—
116	test_greedy_mods_order/order/1	core	verified	optimized · mods	x y z	6	0.0535	0.0307	0.613	0.087
117	test_pentabox_exd/order/1	core	verified	optimized · mods	x4 x7 x6 x5 x3 x2 x1 x0 · H=x4	7	0.0537	0.0291	*not run*	—
118	finiteA_4_278_5_0__0_2_1_0_1_0_0_0_1/symbolic	symbolic	verified	monotone · mods	s x4 x2 x8 x1 · H=x4	5	0.0537	0.0291	1.22	0.044
119	finiteE_3_13__3_0__1_0_1_1/symbolic	symbolic	verified	monotone · mods	s x0 x2 x3 · H=x0	4	0.0539	0.0281	1.21	0.044
120	finiteCx12_6_215_11_0__6_1_1_0_1_0_1_1_0/fixed	fixed	verified	monotone · mods	x4 x2 x1 x0 x7 x6 · H=x4	1	0.0539	0.0279	5.43	0.0099
121	finiteE_1_1__3_0__3_0_0_0/fixed	fixed	verified	monotone · mods	x0 · H=x0	1	0.0540	0.0281	1.22	0.044
122	finiteE_3_11__3_0__1_1_0_1/symbolic	symbolic	verified	monotone · mods	s t x0 x3 x1 · H=x0	8	0.0543	0.0292	1.42	0.038
123	finiteAx12_4_312_5_0__0_0_0_2_1_1_0_0_1/symbolic	symbolic	verified	monotone · mods	t s x4 x5 x3 x8 · H=x4	9	0.0543	0.0306	1.22	0.044
124	test_gluza_ex30/order/1	core	verified	linear_negative · given	x2 x6 x7 x5 x1 x4 x0	1	0.0544	0.0309	*not run*	—
125	test_ex411_box/solve/2	core	verified	optimized · mods	x4 x3 x2 x1 s13 s12 · H=x4	1	0.0545	0.0291	0.615	0.089
126	finiteB_4_197_5_0__2_0_1_0_0_0_1_1_0/symbolic	symbolic	verified	monotone · mods	s x0 x7 x6 x2 · H=x0	13	0.0546	0.0291	1.42	0.038
127	finiteE_2_5__3_0__2_0_1_0/symbolic	symbolic	verified	monotone · mods	s x0 x2 · H=x0	3	0.0547	0.0297	1.22	0.045
128	test_x2_minus_y/solve/1	core	verified	optimized · mods	x y	1	0.0549	0.0309	0.614	0.089
129	test_greedy_sotd_order/order/3	core	verified	optimized · mods	y x	9	0.0551	0.0296	0.613	0.090
130	finiteA_5_155_6_0__2_1_0_1_1_0_0_1_0/symbolic	symbolic	verified	monotone · mods	s t x4 x3 x1 x0 x7 · H=x4	10†	0.0553	0.0282	2.03	0.027
131	test_massive_triangle/solve/1	core	verified	optimized · mods	x1 x0 s	6	0.0554	0.0304	0.610	0.091
132	finiteB_4_85_5_0__2_0_1_0_1_0_1_0_0/symbolic	symbolic	verified	monotone · mods	s x4 x2 x0 x6 · H=x4	4†	0.0555	0.0297	1.23	0.045
133	finiteE_2_10__3_0__0_2_0_1/symbolic	symbolic	verified	monotone · mods	s t x1 x3 · H=x1	5	0.0560	0.0299	1.22	0.046
134	finiteE_2_9__3_0__2_0_0_1/symbolic	symbolic	verified	monotone · mods	s x0 x3 · H=x0	1	0.0560	0.0299	1.22	0.046

#	Problem	Scope	Status	Method · ordering	Order (H = anchor)	Cells	Solve s	Verify s	Native s	Ratio
135	finiteA_4_29_5_0__2_0_1_1_1_0_0_0_0/symbolic	symbolic	verified	monotone · mods	s x4 x2 x0 x3 · H=x4	6	0.0561	0.0298	1.22	0.046
136	test_ex44_B1_B4/solve/1	core	verified	optimized · mods	z y x	0	0.0563	0.0288	0.615	0.092
137	test_greedy_sotd_order/order/4	core	verified	optimized · mods	x y	7	0.0564	0.0324	0.610	0.092
138	test_hypercube_5/solve/1	core	verified	optimized · mods	z y x	2	0.0571	0.0293	0.610	0.094
139	test_ex422_bubble/solve/2	core	verified	optimized · mods	x1 x2 b · H=x1	2	0.0571	0.0297	0.611	0.094
140	finiteA_5_121_6_0__2_0_0_1_1_1_1_0_0/symbolic	symbolic	verified	monotone · mods	s x4 x6 x3 x5 x0 · H=x4	6†	0.0572	0.0282	1.23	0.047
141	finiteAx12_3_296_4_0__0_0_0_2_0_1_0_0_1/symbolic	symbolic	verified	monotone · mods	t s x3 x5 x8 · H=x3	6	0.0581	0.0303	1.22	0.048
142	test_ex412_box/solve/2	core	verified	optimized · mods	x4 x1 p1s s13 x3 x2 s12	1	0.0581	0.0295	0.614	0.095
143	finiteC_5_214_10_0__0_6_1_0_1_0_1_1_0/fixed	fixed	verified	monotone · mods	x4 x6 x7 x2 x1 · H=x4	1	0.0584	0.0286	1.63	0.036
144	test_pentabox_exb/solve/3	core	verified	optimized · mods	x4 x7 x6 x5 x3 x2 x1 x0 · H=x4	3	0.0592	0.0308	1.01	0.059
145	test_ex411_box/solve/1	core	verified	optimized · mods	x4 x3 x2 x1 s13 s12	1	0.0593	0.0297	0.610	0.097
146	test_ex422_bubble/solve/1	core	verified	optimized · mods	x1 x2 b · H=x1	2	0.0598	0.0301	0.614	0.097
147	finiteB_3_21_4_0__2_0_1_0_1_0_0_0_0/symbolic	symbolic	verified	monotone · mods	s x4 x2 x0 · H=x4	3	0.0606	0.0289	1.22	0.050
148	test_gluza_ex33_trivial/solve/3	core	verified	optimized · mods	x4 x7 x6 x5 x3 x2 x1 x0 · H=x4	1	0.0615	0.0305	*not run*	—
149	test_massive_triangle/solve/2	core	verified	optimized · mods	x1 x0 s	4	0.0616	0.0309	0.811	0.076
150	test_hypercube_2/solve/1	core	verified	optimized · mods	y x z	1	0.0616	0.0289	0.605	0.10
151	test_greedy_sotd_order/solve/1	core	verified	optimized · mods	y x	9	0.0618	0.0301	0.610	0.10
152	test_unbounded_neg/solve/1	core	verified	optimized · mods	x	1	0.0628	0.0317	0.608	0.10
153	test_input_unmodified/order/1	core	verified	optimized · mods	t u z y x	2	0.0630	0.0288	0.605	0.10
154	finiteA_6_125_7_0__2_0_1_1_1_1_1_0_0/fixed	fixed	verified	monotone · mods	x4 x6 x2 x0 x5 x3 · H=x4	23†	0.0631	0.0295	1.82	0.035
155	test_input_unmodified/order/3	core	verified	optimized · mods	t u z y x	2	0.0639	0.0287	0.611	0.10
156	test_unbounded/solve/1	core	verified	optimized · mods	x · H=x	1	0.0642	0.0310	0.608	0.11
157	test_hypercube_4/solve/1	core	verified	optimized · mods	z y x	1	0.0644	0.0298	0.608	0.11
158	finiteAx12_6_377_10_0__5_0_0_1_1_1_1_0_1/fixed	fixed	verified	monotone · mods	x4 x6 x0 x8 x5 x3 · H=x4	15†	0.0644	0.0295	2.42	0.027
159	finiteAx12_6_159_7_0__2_1_1_1_1_0_0_1_0/fixed	fixed	verified	monotone · mods	x4 x2 x0 x3 x1 x7 · H=x4	8†	0.0655	0.0283	1.62	0.040
160	test_hypercube_3/solve/1	core	verified	optimized · mods	z y x	1	0.0657	0.0308	0.608	0.11
161	test_no_real_roots/solve/1	core	verified	optimized · mods	x	1	0.0658	0.0345	0.605	0.11
162	test_input_unmodified/solve/4	core	verified	optimized · mods	x z y u t	2	0.0660	0.0308	0.816	0.081
163	finiteAx12_6_317_7_0__2_0_1_1_1_1_0_0_1/fixed	fixed	verified	monotone · mods	x4 x2 x0 x8 x3 x5 · H=x4	32†	0.0661	0.0297	1.82	0.036
164	finiteA_6_414_10_0__0_5_1_1_1_0_0_1_1/fixed	fixed	verified	monotone · mods	x4 x2 x8 x3 x1 x7 · H=x4	7	0.0662	0.0299	5.43	0.012
165	finiteA_5_93_6_0__2_0_1_1_1_0_1_0_0/fixed	fixed	verified	monotone · mods	x4 x6 x2 x0 x3 · H=x4	19†	0.0664	0.0293	1.21	0.055
166	test_cylinder_3d/solve/1	core	verified	optimized · mods	z y x	1	0.0664	0.0312	0.610	0.11
167	finiteAx12_3_26_4_0__0_2_0_1_1_0_0_0_0/symbolic	symbolic	verified	monotone · mods	t s x4 x3 x1 · H=x4	8	0.0666	0.0310	1.43	0.047
168	finiteB_5_453_6_0__2_0_1_0_0_0_1_1_1/fixed	fixed	verified	monotone · mods	x0 x8 x7 x6 x2 · H=x0	77	0.0667	0.0278	2.02	0.033
169	test_greedy_sotd_order/order/5	core	verified	optimized · mods	y x	9	0.0672	0.0350	0.615	0.11
170	test_input_unmodified/order/2	core	verified	optimized · mods	t u z y x	2	0.0673	0.0307	0.615	0.11
171	finiteA_6_159_7_0__2_1_1_1_1_0_0_1_0/fixed	fixed	verified	monotone · mods	x4 x2 x0 x3 x1 x7 · H=x4	8†	0.0681	0.0290	1.61	0.042
172	finiteA_5_309_6_0__2_0_1_0_1_1_0_0_1/fixed	fixed	verified	monotone · mods	x4 x2 x0 x8 x5 · H=x4	25†	0.0681	0.0310	1.42	0.048
173	finiteAx12_7_446_11_0__0_5_1_1_1_1_0_1_1/fixed	fixed	verified	monotone · mods	x4 x2 x8 x7 x5 x3 x1 · H=x4	8	0.0682	0.0483	timeout	—
174	finiteA_4_282_5_0__0_2_0_1_1_0_0_0_1/symbolic	symbolic	verified	monotone · mods	s t x4 x3 x1 x8 · H=x4	36	0.0686	0.0300	1.22	0.056
175	finiteA_6_317_7_0__2_0_1_1_1_1_1_0_0_1/fixed	fixed	verified	monotone · mods	x4 x2 x0 x8 x3 x5 · H=x4	32†	0.0688	0.0305	2.22	0.031
176	finiteA_6_377_10_0__5_0_0_1_1_1_1_0_1/fixed	fixed	verified	monotone · mods	x4 x6 x0 x8 x5 x3 · H=x4	15†	0.0689	0.0297	2.62	0.026
177	finiteAx12_4_170_5_0__0_2_0_1_0_1_0_1_0/symbolic	symbolic	verified	monotone · mods	t s x1 x7 x5 x3 · H=x1	151	0.0690	0.0297	1.23	0.056
178	test_x_y_z_minus_1/solve/1	core	verified	optimized · mods	x y z	2	0.0691	0.0320	0.609	0.11
179	finiteB_6_343_7_0__2_1_1_0_1_0_1_0_1/fixed	fixed	verified	monotone · mods	x4 x8 x6 x2 x0 x1 · H=x4	15	0.0693	0.0298	1.42	0.049
180	test_ex44_B1_B2/solve/1	core	verified	optimized · mods	z y x	1	0.0693	0.0304	0.605	0.11
181	finiteA_7_446_11_0__0_5_1_1_1_1_0_1_1/fixed	fixed	verified	monotone · mods	x4 x2 x8 x7 x5 x3 x1 · H=x4	8	0.0702	0.0305	timeout	—
182	finiteCx12_6_476_10_0__0_0_5_1_1_0_1_1_1/fixed	fixed	verified	monotone · mods	x4 x6 x2 x8 x7 x3 · H=x4	7	0.0703	0.0297	1.62	0.043
183	finiteA_4_85_5_0__2_0_1_0_1_0_1_0_0/symbolic	symbolic	verified	monotone · mods	s x4 x6 x2 x0 · H=x4	15	0.0706	0.0282	1.62	0.043
184	test_ex44_B1_B2_B3/solve/1	core	verified	optimized · mods	z y x	4	0.0709	0.0292	0.615	0.12
185	finiteA_5_87_6_0__2_1_1_0_1_0_1_0_0/symbolic	symbolic	verified	monotone · mods	s x4 x6 x2 x0 x1 · H=x4	19†	0.0719	0.0291	1.43	0.050
186	finiteC_6_476_10_0__0_0_5_1_1_0_1_1_1/fixed	fixed	verified	monotone · mods	x4 x6 x2 x8 x7 x3 · H=x4	7	0.0719	0.0288	1.62	0.044
187	test_ex44_B1_C1/solve/1	core	verified	optimized · mods	z y x	2	0.0720	0.0293	0.615	0.12
188	finiteC_6_470_10_0__0_5_1_0_1_0_1_1_1/fixed	fixed	verified	monotone · mods	x4 x6 x2 x8 x7 x1 · H=x4	7	0.0722	0.0294	1.62	0.045
189	test_cube/solve/1	core	verified	optimized · mods	z y x	1	0.0730	0.0355	0.613	0.12
190	test_ex44_B1_C2/solve/1	core	verified	optimized · mods	z y x	0	0.0733	0.0305	0.609	0.12
191	finiteA_4_297_5_0__2_0_0_1_0_1_0_0_1/symbolic	symbolic	verified	monotone · mods	s t x0 x5 x3 x8 · H=x0	9	0.0735	0.0293	1.23	0.060

#	Problem	Scope	Status	Method · ordering	Order (H = anchor)	Cells	Solve s	Verify s	Native s	Ratio
192	test_ex44_T_C1/solve/1	core	verified	optimized · mods	z y x	3	0.0736	0.0301	0.610	0.12
193	test_ex44_T_B2/solve/1	core	verified	optimized · mods	z y x	31	0.0737	0.0294	0.613	0.12
194	test_boundary/solve/1	core	verified	optimized · mods	x	0	0.0741	0.0294	0.614	0.12
195	test_hypercube_1/solve/1	core	verified	optimized · mods	z y x	1	0.0743	0.0291	0.609	0.12
196	finiteB_5_279_6_0__2_1_1_0_1_0_0_0_1/symbolic	symbolic	verified	monotone · mods	t s x4 x8 x2 x1 x0 · H=x4	18	0.0744	0.0299	1.62	0.046
197	finiteBx12_6_343_7_0__2_1_1_0_1_0_1_0_1/fixed	fixed	verified	monotone · mods	x4 x8 x6 x2 x0 x1 · H=x4	15	0.0745	0.0296	1.62	0.046
198	finiteAx12_4_297_5_0__2_0_0_1_0_1_0_0_1/symbolic	symbolic	verified	monotone · mods	t s x0 x5 x3 x8 · H=x0	9	0.0746	0.0302	1.21	0.061
199	test_melih_box/solve/3	core	verified	optimized · mods	x4 x6 x5 x3 x7 x2 x1 · H=x4	3	0.0750	0.0312	0.605	0.12
200	test_empty_3d/solve/1	core	verified	optimized · mods	z y x	0	0.0754	0.0334	0.605	0.12
201	finiteB_5_213_6_0__2_0_1_0_1_0_1_1_0/symbolic	symbolic	verified	monotone · mods	s x4 x7 x6 x2 x0 · H=x4	10	0.0764	0.0291	1.82	0.042
202	finiteBx12_5_279_6_0__2_1_1_0_1_0_0_0_1/symbolic	symbolic	verified	monotone · mods	s t x4 x8 x2 x1 x0 · H=x4	18	0.0768	0.0293	1.23	0.063
203	test_triangle2l_split/solve/3	core	verified	optimized · mods	x4 x3 x5 x2 x1 x0 · H=x4	6	0.0769	0.0294	0.614	0.13
204	finiteA_5_313_6_0__2_0_0_1_1_1_0_0_1/symbolic	symbolic	verified	monotone · mods	s t x4 x5 x3 x8 x0 · H=x4	10	0.0770	0.0303	1.62	0.048
205	finiteB_5_341_6_0__2_0_1_0_1_0_1_0_1/symbolic	symbolic	verified	monotone · mods	s x4 x8 x6 x2 x0 · H=x4	11	0.0771	0.0294	1.42	0.054
206	test_pentabox_exb_symb/order/1	core	verified	optimized · mods	s15 s45 s34 s23 s12 x7 x6 x5 x4 x3 x2 x1 x0	5	0.0772	0.0295	*not run*	—
207	test_ex413_pentagon/solve/1	core	verified	optimized · mods	x3 x5 x4 x1 s51 s45 x2 s34 s23 s12	5	0.0775	0.0301	0.608	0.13
208	test_pentabox_exd/solve/3	core	verified	optimized · mods	x4 x7 x6 x5 x3 x2 x1 x0 · H=x4	4	0.0779	0.0311	0.810	0.096
209	finiteA_5_93_6_0__2_0_1_1_1_0_1_0_0/symbolic	symbolic	verified	monotone · mods	s x4 x6 x2 x0 x3 · H=x4	19†	0.0782	0.0284	1.62	0.048
210	test_ex413_pentagon/solve/2	core	verified	optimized · mods	x3 x5 x4 x1 s51 s45 x2 s34 s23 s12	3	0.0783	0.0306	0.615	0.13
211	finiteA_5_157_6_0__2_0_1_1_1_0_0_1_0/symbolic	symbolic	verified	monotone · mods	s x4 x2 x0 x3 x7 · H=x4	7†	0.0793	0.0287	1.62	0.049
212	test_ex414_bnp6/solve/1	core	verified	optimized · mods	s12 s23 x1 x6 x3 x4 x2 x5	1	0.0816	0.0325	0.616	0.13
213	test_ex414_bnp6/solve/2	core	verified	optimized · mods	s12 s23 x1 x6 x3 x4 x2 x5	2	0.0818	0.0317	0.614	0.13
214	test_quadrant/solve/1	core	verified	optimized · mods	x y · H=x	1	0.0844	0.0725	0.614	0.14
215	test_massless_planar_elliptic_symb/order/1	core	verified	local · given	ss1 ss2 ss4 ss5 ss12 ss23 ss34 ss56 ss61 ss45 ss123 ss234 ss345 x1 x6 x2 x3 x4 x5 x0 · H=x1	12	0.0851	0.0297	*not run*	—
216	test_gluza_ex33/order/1	core	verified	linear_negative · given	x6 x1 x4 x0 x2 x7 x5 x3	3	0.0852	0.0307	*not run*	—
217	test_ex44_B1_B2_B4/solve/1	core	verified	optimized · mods	z y x	0	0.0862	0.0290	0.616	0.14
218	finiteAx12_7_381_10_0__4_0_1_1_1_1_1_0_1/fixed	fixed	verified	monotone · given	x8 x4 x6 x0 x2 x5 x3 · H=x8	836	0.0864	0.0698	timeout	—
219	finiteB_6_469_7_0__2_0_1_0_1_0_1_1_1/fixed	fixed	verified	monotone · mods	x4 x7 x6 x8 x2 x0 · H=x4	184	0.0865	0.0498	1.62	0.053
220	finiteCx12_7_478_10_0__0_4_1_1_1_0_1_1_1/fixed	fixed	verified	monotone · mods	x4 x6 x2 x8 x7 x3 x1 · H=x4	8	0.0885	0.0303	timeout	—
221	finiteD_6_219_7_0__2_1_0_1_1_0_1_1_0/fixed	fixed	verified	monotone · mods	x4 x3 x7 x6 x1 x0 · H=x4	262	0.0887	0.0499	1.82	0.049
222	finiteA_4_170_5_0__0_2_0_1_0_1_0_1_0/symbolic	symbolic	verified	monotone · mods	s t x1 x7 x5 x3 · H=x1	151	0.0891	0.0491	1.63	0.055
223	finiteA_3_21_4_0__2_0_1_0_1_0_0_0_0/fixed	fixed	verified	monotone · mods	x4 x2 x0 · H=x4	5	0.0899	0.0308	1.43	0.063
224	finiteA_2_17_3_0__2_0_0_0_1_0_0_0_0/fixed	fixed	verified	monotone · mods	x4 x0 · H=x4	1	0.0901	0.0297	1.22	0.074
225	finiteE_1_1__3_0__3_0_0_0/symbolic	symbolic	verified	monotone · mods	s x0 · H=x0	1	0.0903	0.0303	1.22	0.074
226	finiteC_7_478_10_0__0_4_1_1_1_0_1_1_1/fixed	fixed	verified	monotone · mods	x4 x6 x2 x8 x7 x3 x1 · H=x4	8	0.0908	0.0298	timeout	—
227	finiteAx12_5_314_6_0__0_2_0_1_1_1_0_0_1/symbolic	symbolic	verified	monotone · mods	t s x4 x3 x1 x5 x8 · H=x4	144	0.0918	0.0488	2.02	0.046
228	finiteD_6_373_7_0__2_0_1_0_1_1_1_0_1/fixed	fixed	verified	monotone · mods	x4 x6 x8 x5 x2 x0 · H=x4	141†	0.0932	0.0279	2.62	0.036
229	finiteA_5_316_10_0__0_0_6_1_1_1_0_0_1/symbolic	symbolic	verified	monotone · mods	s t x4 x3 x5 x2 x8 · H=x4	151	0.0937	0.0302	timeout	—
230	finiteAx12_5_316_10_0__0_0_6_1_1_1_0_0_1/symbolic	symbolic	verified	monotone · mods	t s x4 x3 x5 x2 x8 · H=x4	249	0.0945	0.0518	timeout	—
231	finiteB_5_453_6_0__2_0_1_0_0_0_1_1_1/symbolic	symbolic	verified	monotone · mods	s x0 x8 x7 x6 x2 · H=x0	524	0.0945	0.0504	1.43	0.066
232	finiteAx12_5_313_6_0__2_0_0_1_1_1_0_0_1/symbolic	symbolic	verified	monotone · mods	t s x4 x5 x3 x8 x0 · H=x4	10	0.0957	0.0297	1.63	0.059
233	finiteE_4_15__4_0__1_1_1_1/symbolic	symbolic	verified	monotone · mods	s t x0 x3 x2 x1 · H=x0	406	0.104	0.0497	1.22	0.085
234	test_pentabox_exb/order/1	core	verified	optimized · mods	x4 x7 x6 x5 x3 x2 x1 x0 · H=x4	7	0.109	0.0310	*not run*	—
235	test_ex415_bnp7/solve/2	core	verified	optimized · mods	s12 s23 x1 x3 x2 x4 x5 x6 x7	10	0.110	0.0303	0.611	0.18
236	finiteA_7_381_10_0__4_0_1_1_1_1_1_0_1/fixed	fixed	verified	monotone · given	x8 x4 x6 x0 x2 x5 x3 · H=x8	836	0.110	0.0710	timeout	—
237	finiteA_6_125_7_0__2_0_1_1_1_1_1_0_0/symbolic	symbolic	verified	monotone · mods	s x4 x6 x2 x0 x5 x3 · H=x4	23†	0.116	0.0310	1.62	0.071
238	test_ex415_bnp7/solve/1	core	verified	optimized · mods	s12 s23 x1 x3 x2 x4 x5 x6 x7	18	0.118	0.0332	0.605	0.20
239	test_pentabox_exd_symb/order/1	core	verified	optimized · mods	s15 s45 s23 s34 s12 x7 x6 x5 x4 x3 x2 x1 x0	152	0.118	0.0516	*not run*	—
240	finiteB_7_471_13_0__7_1_1_0_1_0_1_1_1/fixed	fixed	verified	monotone · mods	x4 x7 x6 x8 x2 x1 x0 · H=x4	245	0.119	0.0500	1.62	0.074
241	finiteD_7_375_13_0__7_1_1_0_1_1_1_0_1/fixed	fixed	verified	monotone · mods	x4 x6 x8 x5 x2 x0 x1 · H=x4	196†	0.126	0.0294	6.41	0.020
242	failing_test_ex44_T_C1_B2/solve/1	core	verified	optimized · mods	z y x	57	0.126	0.0711	0.811	0.16
243	finiteF_4_15__4_0__1_1_1_1/symbolic	symbolic	verified	monotone · mods	t s x0 x3 x2 x1 · H=x0	348†	0.128	0.0700	1.63	0.078
244	test_gluza_ex33_trivial/order/1	core	verified	optimized · mods	x4 x7 x6 x5 x3 x2 x1 x0 · H=x4	0	0.129	0.0327	*not run*	—
245	failing_test_ex43/project/1	core	verified	optimized · mods	t z y x	10	0.131	0.0497	*not run*	—
246	finiteBx12_7_471_13_0__7_1_1_0_1_0_1_1_1/fixed	fixed	verified	monotone · mods	x4 x7 x6 x8 x2 x1 x0 · H=x4	245	0.132	0.0500	1.62	0.081
247	finiteA_5_314_6_0__0_2_0_1_1_1_0_0_1/symbolic	symbolic	verified	monotone · mods	s t x4 x3 x1 x5 x8 · H=x4	144	0.138	0.0492	2.42	0.057
248	finiteD_7_251_12_0__6_1_0_1_1_1_1_1_0/fixed	fixed	verified	monotone · mods	x4 x3 x7 x6 x1 x0 x5 · H=x4	364	0.140	0.0493	122	0.0012

#	Problem	Scope	Status	Method · ordering	Order (H = anchor)	Cells	Solve s	Verify s	Native s	Ratio
249	finiteC_5_227_6_0__2_1_0_0_0_1_1_1_0/symbolic	symbolic	verified	monotone · mods	t s x0 x6 x5 x1 x7 · H=x0	354†	0.148	0.0702	14.7	0.010
250	finiteB_6_469_7_0__2_0_1_0_1_0_1_1_1/symbolic	symbolic	verified	monotone · mods	s x4 x7 x6 x8 x2 x0 · H=x4	184	0.155	0.0496	2.82	0.055
251	finiteC_5_214_10_0__0_6_1_0_1_0_1_1_0/symbolic	symbolic	verified	monotone · mods	t s x4 x6 x2 x1 x7 · H=x4	354†	0.157	0.0920	timeout	—
252	finiteA_5_412_6_0__0_0_2_1_1_0_0_1_1/symbolic	symbolic	verified	monotone · mods	s x4 x2 x8 x7 x3 · H=x4	6	0.159	0.0292	2.02	0.078
253	finiteA_5_309_6_0__2_0_1_0_1_1_0_0_1/symbolic	symbolic	verified	monotone · mods	s x4 x2 x0 x8 x5 · H=x4	46†	0.159	0.0307	1.81	0.088
254	finiteB_5_339_6_0__2_1_0_0_1_0_1_0_1/symbolic	symbolic	verified	monotone · mods	t s x4 x8 x6 x1 x0 · H=x4	1,079	0.159	0.132	1.62	0.098
255	finiteBx12_5_339_6_0__2_1_0_0_1_0_1_0_1/symbolic	symbolic	verified	monotone · mods	s t x4 x8 x6 x1 x0 · H=x4	954	0.161	0.111	1.62	0.100
256	finiteC_5_412_6_0__0_0_2_1_1_0_0_1_1/symbolic	symbolic	verified	monotone · mods	t s x4 x3 x2 x8 x7 · H=x4	228	0.179	0.0693	1.63	0.11
257	test_elliptic2l_physical_symb/solve/2	core	verified	optimized · mods	x4 x0 x1 x3 x6 pp4 x5 s t x2 · H=x4	15†	0.196	0.0304	*not run*	—
258	finiteAx12_5_410_10_0__0_6_0_1_1_0_0_1_1/symbolic	symbolic	verified	monotone · mods	t s x4 x7 x3 x1 x8 · H=x4	973	0.200	0.112	2.43	0.082
259	test_melih_box/order/1	core	verified	optimized · mods	x4 x6 x5 x3 x7 x2 x1 · H=x4	1	0.205	0.0299	*not run*	—
260	finiteA_5_345_6_0__2_0_0_1_1_0_1_0_1/symbolic	symbolic	verified	monotone · mods	s x4 x6 x0 x8 x3 · H=x4	12†	0.218	0.0301	2.02	0.11
261	finiteD_6_219_7_0__2_1_0_1_1_0_1_1_0/symbolic	symbolic	verified	monotone · mods	s x4 x3 x7 x6 x1 x0 · H=x4	262	0.221	0.0490	6.24	0.035
262	finiteB_6_343_7_0__2_1_1_0_1_0_1_0_1/symbolic	symbolic	verified	monotone · mods	t s x4 x8 x6 x2 x1 x0 · H=x4	2,249	0.222	0.233	1.62	0.14
263	test_ex44_HB1_HB2_HB3/solve/1	core	verified	optimized · mods	z y x	0	0.227	0.111	0.615	0.37
264	finiteBx12_6_343_7_0__2_1_1_0_1_0_1_0_1/symbolic	symbolic	verified	monotone · mods	s t x4 x8 x6 x2 x1 x0 · H=x4	1,963	0.242	0.192	1.62	0.15
265	finiteAx12_6_190_7_0__0_2_1_1_1_1_0_1_0/symbolic	symbolic	verified	monotone · mods	t s x4 x7 x5 x3 x2 x1 · H=x4	130	0.255	0.0503	1.62	0.16
266	finiteAx12_4_282_5_0__0_2_0_1_1_0_0_0_1/symbolic	symbolic	verified	monotone · mods	t s x4 x3 x1 x8 · H=x4	36	0.257	0.0287	1.23	0.21
267	test_ex44_HT_C1_HB2/solve/1	core	verified	optimized · mods	z y x	0	0.269	0.253	1.01	0.27
268	finiteD_6_373_7_0__2_0_1_0_1_1_1_0_1/symbolic	symbolic	verified	monotone · mods	s x4 x8 x6 x5 x2 x0 · H=x4	3,096	0.276	0.315	*not run*	—
269	test_ex44_HT_HB2_HB3/solve/1	core	verified	optimized · mods	y x z	0	0.280	0.0711	0.811	0.35
270	finiteCx12_5_158_6_0__0_2_1_1_1_0_0_1_0/symbolic	symbolic	verified	optimized · mods	t s x3 x2 x1 x4 x7	1,273	0.308	0.211	2.42	0.13
271	test_elliptic2l_physical/order/1	core	verified	optimized · mods	x4 x6 x3 x1 x0 x5 x2 · H=x4	1,196	0.316	0.295	3.02	0.10
272	finiteAx12_5_376_6_0__0_0_0_2_1_1_1_0_1/symbolic	symbolic	verified	monotone · mods	t s x4 x6 x5 x3 x8 · H=x4	1,744	0.331	0.253	25.5	0.013
273	finiteA_5_286_6_0__0_2_1_1_1_0_0_0_1/symbolic	symbolic	verified	optimized · mods	s t x3 x2 x1 x4 x8	3,709	0.338	0.334	4.43	0.076
274	finiteA_6_190_7_0__0_2_1_1_1_1_1_0_1/symbolic	symbolic	verified	monotone · mods	s t x4 x7 x5 x3 x2 x1 · H=x4	130	0.363	0.0702	2.02	0.18
275	test_elliptic2l_physical/order/2	core	verified	optimized · mods	x4 x6 x3 x1 x0 x5 x2 · H=x4	1,196	0.364	0.294	2.61	0.14
276	finiteAx12_5_286_6_0__0_2_1_1_1_0_0_0_1/symbolic	symbolic	verified	optimized · mods	t s x3 x2 x1 x4 x8	5,489	0.410	0.455	6.23	0.066
277	finiteC_5_158_6_0__0_2_1_1_1_0_0_1_0/symbolic	symbolic	verified	optimized · mods	t s x3 x2 x1 x4 x7	2,197	0.416	0.354	3.27	0.13
278	finiteA_5_410_10_0__0_6_0_1_1_0_0_1_1/symbolic	symbolic	verified	monotone · mods	s t x4 x7 x3 x1 x8 · H=x4	973	0.435	0.132	4.02	0.11
279	finiteAx12_5_301_6_0__2_0_1_1_0_1_0_0_1/symbolic	symbolic	verified	monotone · mods	t s x0 x3 x2 x5 x8 · H=x0	5,462	0.436	0.617	6.41	0.068
280	finiteA_5_301_6_0__2_0_1_1_0_1_0_0_1/symbolic	symbolic	verified	monotone · mods	s t x0 x3 x2 x5 x8 · H=x0	7,929	0.556	0.943	6.84	0.081
281	test_elliptic2l_physical/solve/3	core	verified	optimized · mods	x4 x0 x1 x3 x6 x5 x2 · H=x4	1,193	0.615	0.545	0.810	0.76
282	test_elliptic2l_physical/solve/4	core	verified	optimized · mods	x4 x0 x1 x3 x6 x5 x2 · H=x4	3,254	0.713	0.619	0.810	0.88
283	finiteAx12_5_171_6_0__2_1_0_1_0_1_0_1_0/symbolic	symbolic	verified	monotone · mods	t s x0 x7 x5 x3 x1 · H=x0	15,730	0.839	1.59	1.62	0.52
284	finiteA_5_310_6_0__0_2_1_0_1_1_0_0_1/symbolic	symbolic	verified	monotone · mods	s x4 x2 x8 x5 x1 · H=x4	6	1.01	0.0305	1.41	0.71
285	finiteA_5_376_6_0__0_0_0_2_1_1_1_0_1/symbolic	symbolic	verified	monotone · mods	s t x4 x6 x5 x3 x8 · H=x4	12,357	1.05	1.94	30.5	0.035
286	finiteAx12_5_31_6_0__2_1_1_1_1_0_0_0_0/symbolic	symbolic	verified	monotone · mods	t s x4 x3 x2 x1 x0 · H=x4	35,354	1.83	4.01	5.83	0.31
287	finiteAx12_6_159_7_0__2_1_1_1_1_0_0_1_0/symbolic	symbolic	verified	monotone · mods	t s x4 x3 x2 x1 x0 x7 · H=x4	40,697†	2.11	4.37	memory	—
288	test_ex_e5/order/1	core	verified	optimized · mods	p1s x4 x5 x3 x2 x1 x0 · H=x4	0	2.39	5.70	*not run*	—
289	test_ex_e5/solve/3	core	verified	optimized · mods	p1s x4 x5 x3 x2 x1 x0 · H=x4	11,968†	2.55	5.74	5.83	0.44
290	finiteA_5_171_6_0__2_1_0_1_0_1_0_1_0/symbolic	symbolic	verified	monotone · mods	s t x0 x7 x5 x3 x1 · H=x0	15,730	3.30	4.14	1.62	2.0
291	test_ex_e5/solve/4	core	verified	optimized · mods	p1s x4 x5 x3 x2 x1 x0 · H=x4	16,232	3.55	9.63	6.83	0.52
292	finiteA_5_31_6_0__2_1_1_1_1_0_0_0_0/symbolic	symbolic	verified	monotone · mods	s t x4 x3 x2 x1 x0 · H=x4	74,351	3.70	8.29	4.43	0.84
293	test_ex_e5/solve/5	core	verified	optimized · mods	p1s x4 x5 x3 x2 x1 x0 · H=x4	54,026	4.06	10.4	10.6	0.38
294	finiteA_6_159_7_0__2_1_1_1_1_0_0_1_0/symbolic	symbolic	verified	monotone · mods	s t x4 x3 x2 x1 x0 x7 · H=x4	85,868†	4.23	9.50	memory	—
295	test_massless_planar_elliptic/solve/3	core	verified	baseline · given	x4 x3 x2 x6 x5 x1 x0	138,678	8.76	23.0	17.7	0.50
296	test_massless_planar_elliptic/order/1	core	verified	baseline · given	x4 x3 x2 x6 x5 x1 x0	131,072	8.77	22.7	*not run*	—
297	finiteC_6_438_7_0__0_2_1_0_1_1_0_1_1/symbolic	symbolic	verified	monotone · mods	t s x4 x5 x2 x1 x8 x7 · H=x4	115,955†	8.78	19.2	37.9	0.23
298	test_gluza_ex30/solve/3	core	verified	bilinear_positive/local · given	x1 x7 x2 x6 x5 x4 x0	80,980	21.0	16.0	*not run*	—
299	finiteC_5_31_6_0__2_1_1_1_1_0_0_0_0/symbolic	symbolic	verified	monotone · mods	t s x4 x3 x2 x1 x0 · H=x4	271,030	21.5	51.2	2.62	8.2
300	finiteC_6_95_7_0__2_1_1_1_1_0_1_0_0/symbolic	symbolic	verified	monotone · mods	t s x4 x3 x2 x1 x0 x6 · H=x4	276,039†	22.9	52.9	timeout	—
301	hanging_test_ex38/solve/1	core	verified	optimized · mods	t z y x	124,744†	29.8	61.2	18.9	1.6
302	test_elliptic2l_physical_symb/order/1	core	verified	monotone · given	pp4 t s x6 x2 x3 x1 x0 x5 x4 · H=x6	269,061	30.3	70.9	*not run*	—
303	finiteCx12_6_438_7_0__0_2_1_0_1_1_0_1_1/symbolic	symbolic	verified	monotone · mods	s t x4 x5 x2 x1 x8 x7 · H=x4	115,955†	62.7	37.3	58.8	1.1
304	finiteA_6_317_7_0__2_0_1_1_1_1_0_0_1/symbolic	symbolic	**unresolved**	local · given	s t x4 x0 x2 x3 x5 x8 · H=x4	—	> 3600[w]	—	timeout	—
305	finiteA_6_318_7_0__0_2_1_1_1_1_0_0_1/symbolic	symbolic	**unresolved**	local · given	s t x4 x1 x2 x3 x5 x8 · H=x4	—	> 3600[w]	—	timeout	—

#	Problem	Scope	Status	Method · ordering	Order (H = anchor)	Cells	Solve s	Verify s	Native s	Ratio
306	finiteA_6_377_10_0__5_0_0_1_1_1_1_0_1/symbolic	symbolic	**unresolved**	local · given	s t x4 x0 x3 x5 x6 x8 · H=x4	—	> 3600[w]	—	timeout	—
307	finiteA_6_414_10_0__0_5_1_1_1_0_0_1_1/symbolic	symbolic	**unresolved**	local · mcts	s t x8 x1 x2 x3 x4 x7 · H=x8	—	> 3600[w]	—	timeout	—
308	finiteA_7_381_10_0__4_0_1_1_1_1_1_0_1/symbolic	symbolic	**unresolved**	local · mcts	s t x8 x0 x2 x3 x4 x5 x6 · H=x8	—	> 3600[w]	—	timeout	—
309	finiteA_7_446_11_0__0_5_1_1_1_1_0_1_1/symbolic	symbolic	**unresolved**	local · given	s t x5 x1 x2 x3 x4 x7 x8 · H=x5	—	> 3600[w]	—	timeout	—
310	finiteAx12_6_317_7_0__2_0_1_1_1_1_0_0_1/symbolic	symbolic	**unresolved**	local · mcts	s t x8 x0 x2 x3 x4 x5 · H=x8	—	> 3600[w]	—	timeout	—
311	finiteAx12_6_318_7_0__0_2_1_1_1_1_0_0_1/symbolic	symbolic	**unresolved**	local · mcts	s t x8 x1 x2 x3 x4 x5 · H=x8	—	> 3600[w]	—	timeout	—
312	finiteAx12_6_377_10_0__5_0_0_1_1_1_1_0_1/symbolic	symbolic	**unresolved**	local · mcts	s t x8 x0 x3 x4 x5 x6 · H=x8	—	> 3604[w]	—	timeout	—
313	finiteAx12_6_414_10_0__0_5_1_1_1_0_0_1_1/symbolic	symbolic	**unresolved**	local · mcts	s t x8 x1 x2 x3 x4 x7 · H=x8	—	> 3600[w]	—	timeout	—
314	finiteAx12_7_381_10_0__4_0_1_1_1_1_1_0_1/symbolic	symbolic	**unresolved**	local · mcts	s t x8 x0 x2 x3 x4 x5 x6 · H=x8	—	> 3600[w]	—	timeout	—
315	finiteAx12_7_446_11_0__0_5_1_1_1_1_0_1_1/symbolic	symbolic	**unresolved**	local · mcts	s t x8 x1 x2 x3 x4 x5 x7 · H=x8	—	> 3600[w]	—	timeout	—
316	finiteB_7_471_13_0__7_1_1_0_1_0_1_1_1/symbolic	symbolic	**unresolved**	local · mcts	s t x0 x6 x7 x2 x1 x4 x8 · H=x0	—	> 3600[w]	—	timeout	—
317	finiteBx12_7_471_13_0__7_1_1_0_1_0_1_1_1/symbolic	symbolic	**unresolved**	local · mcts	s t x0 x1 x2 x4 x6 x7 x8 · H=x0	—	> 3600[w]	—	timeout	—
318	finiteC_6_215_11_0__6_1_1_0_1_0_1_1_0/symbolic	symbolic	**unresolved**	local · mcts	s t x7 x0 x1 x2 x4 x6 · H=x7	—	> 3600[w]	—	timeout	—
319	finiteC_6_231_7_0__2_1_1_0_0_1_1_1_0/symbolic	symbolic	**unresolved**	local · mcts	s t x7 x0 x1 x2 x5 x6 · H=x7	—	> 3600[w]	—	timeout	—
320	finiteC_6_470_10_0__0_5_1_0_1_0_1_1_1/symbolic	symbolic	**unresolved**	local · mcts	s t x7 x1 x2 x4 x6 x8 · H=x7	—	> 3600[w]	—	timeout	—
321	finiteC_6_476_10_0__0_0_5_1_1_0_1_1_1/symbolic	symbolic	**unresolved**	local · mcts	s t x7 x2 x3 x4 x6 x8 · H=x7	—	> 3600[w]	—	timeout	—
322	finiteC_7_247_10_0__4_1_1_0_1_1_1_1_0/symbolic	symbolic	**unresolved**	local · mcts	s t x7 x0 x1 x2 x4 x5 x6 · H=x7	—	> 3600[w]	—	*not run*	—
323	finiteC_7_478_10_0__0_4_1_1_1_0_1_1_1/symbolic	symbolic	**unresolved**	local · mcts	s t x7 x1 x2 x3 x4 x6 x8 · H=x7	—	> 3600[w]	—	*not run*	—
324	finiteCx12_6_215_11_0__6_1_1_0_1_0_1_1_0/symbolic	symbolic	**unresolved**	local · mcts	s t x7 x0 x1 x2 x4 x6 · H=x7	—	> 3600[w]	—	*not run*	—
325	finiteCx12_6_231_7_0__2_1_1_0_0_1_1_1_0/symbolic	symbolic	**unresolved**	local · mcts	s t x7 x0 x1 x2 x5 x6 · H=x7	—	> 3600[w]	—	*not run*	—
326	finiteCx12_6_476_10_0__0_0_5_1_1_0_1_1_1/symbolic	symbolic	**unresolved**	local · mcts	s t x7 x2 x3 x4 x6 x8 · H=x7	—	> 3600[w]	—	*not run*	—
327	finiteCx12_7_247_10_0__4_1_1_0_1_1_1_1_0/symbolic	symbolic	**unresolved**	local · mcts	s t x7 x0 x1 x2 x4 x5 x6 · H=x7	—	> 3600[w]	—	*not run*	—
328	finiteCx12_7_478_10_0__0_4_1_1_1_0_1_1_1/symbolic	symbolic	**unresolved**	local · mcts	— · H=x7	—	> 3600[h]	—	*not run*	—
329	finiteD_7_251_12_0__6_1_0_1_1_1_1_1_0/symbolic	symbolic	**unresolved**	local · mcts	— · H=x0	—	> 3600[h]	—	*not run*	—
330	finiteD_7_375_13_0__7_1_1_0_1_1_1_0_1/symbolic	symbolic	**unresolved**	local · mcts	— · H=x0	—	> 3600[h]	—	*not run*	—
331	test_gluza_ex33/solve/3	core	**unresolved**	local · mcts	— · H=x3	—	> 7200[h]	—	*not run*	—

Unresolved rows. The **Solve s** column shows the longest attempt, a lower bound and not a solve time: [h] the 2026-10-10 one-hour campaign (Gluza33: 7,200 s), [w] a 2026-10-12 one-hour attempt with improved local-engine recipes and a warm shared operation cache (binary `c8f2c62f`, integration `4dbe440`), [c] a cold attempt of this run. For `mcts` recipes without a fixed order, the order shown is the one the 2026-10-12 attempt was working on. Each attempt's status, wall time, peak memory and binary are in `report-data.json` (`rows[].attempts`).

1.2 Opt-in projective contract

These rows use the user-approved contract extension `solver.projective_sections = true` (Section 2, item X1): a cell's upper boundary may be a positive root that escapes to +∞ inside the cell. They are reported separately and **do not change the strict count of 303 / 331**.

Problem	Engine (base)	Order (H = base anchor)	Cells	Solve s	Verify s	Strict contract	Native s	Sanity check
positive Gluza33 (test_gluza_ex33/solve/3)	bilinear_general (optimized)	x4 x7 x2 x1 x0 x6 x3 x5 · H=x4	2,372	0.49	0.56	unresolved	not run	4,744 fresh points in 2,372 cells, 0 failures
positive Gluza30 (test_gluza_ex30/solve/3)	bilinear_general (optimized)	x1 x7 x2 x6 x5 x4 x0	1	0.11	0.03	80,980 cells, 20.98 s	not run	—

Projective rows: same cold protocol and binary as the strict table (`benchmark/campaign/projective/`). The sanity check (seed 20261010, two fresh rational points per cell, original TOML expressions evaluated exactly) is outside all timings and supplies no proof credit.

2 Improvements over the original GCAD

The reference is Vitaly Magerya's GCAD (Python/SymPy front end over FLINT), which implements Strzeboński's 2000 generic CAD: generic projection (leading coefficients, discriminants and pairwise resultants of a square-free basis), recursive lifting with exact real-root isolation, the MODS/SOTD/T1 ordering heuristics and an aggressive cell merge. Everything listed below goes beyond that.

Vocabulary. A CAD splits the solution set into *cells*. *Projection* eliminates variables one at a time and collects the polynomials (resultants, discriminants, leading coefficients) whose zeros may change the picture; *lifting* goes back up, picking rational *sample* points and isolating real roots above them. The *order* fixes which variable is eliminated first. A *certificate* is an exact, replayable proof (e.g. that a polynomial never vanishes on the domain) that lets a polynomial be dropped. A *guard* is a condition that must hold for a local shortcut to be valid. The *anchor* of a homogeneous problem is the positive variable that is fixed to 1. *Generic* means correct outside an explicit finite list of lower-dimensional exceptional sets.

Tags. default-on active without any setting; opt-in enabled by an option, engine or ordering choice; optional patch optional dependency patch, never in the default build; diagnostic measurement or audit tool, not part of a solve; NEW 2026-10-12 written in the 2026-10-12 session (integrated at commit `9baf91c`: 753 fast tests pass there, against 662 at the `81d9ac5` checkpoint).

Reading the numbers. Unless a line says otherwise, an effect is one measured case or operand: it shows what the change does, not a corpus-wide speedup (corpus-wide timings are in Section 1). "Timeout" results never count as solved. All reductions keep the exact default contract: every cell is proved, every interior `F=0` sheet and same-sign seam stays a boundary. The only exception is the opt-in contract extension X1, which must be switched on explicitly and is never counted in the strict totals.

2.1 Exact algebra and kernels

- A1 **Rust and Symbolica exact backend.** All polynomial arithmetic, GCDs, factorization and resultants run in Rust on Symbolica, in single-threaded workers. *Effect:* in the matched parity audit (release `9025e71`, identical orders, plain `baseline` projection), symGCAD reproduced the raw cell counts of all 483 sign-expanded problems GCAD completes; projection + lifting + merging took a geometric mean of 0.1466 of GCAD's time (6.82× faster), whole processes 0.0620 (16.1×). The current end-to-end comparison is in Section 1. default-on
- A2 **Sign-preserving exact input normalization.** Inequalities become signed positive polynomials; rational content is cleared without flipping a sign. `false`, a strict zero constraint and an identically zero split are told apart, and exponent overflow, undeclared symbols and approximate kinematics are rejected. default-on
- A3 **Full variable maps.** Every polynomial keeps its complete ordered variable map, inactive variables included, so cached algebra can never be reused under the wrong ordering or chart. default-on
- A4 **Deterministic resultants.** Production resultants use Symbolica's deterministic integer Lazard–Ducos routine. Symbolica's automatic dispatcher, whose modular CRT path can stop after random-prime checks, is confined to profiling; caches built with it were invalidated. default-on
- A5 **Closed-form low-degree discriminants.** Exact invariant formulas up to degree 4 replace the generic `Res(p, p')`. *Effect:* captured quartic (642 ⟶ 69,474 terms): 2.71 s / 48 MiB against 29.77 s / 472 MiB (10.98× time, 9.83× memory). default-on
- A6 **Exact evaluation/interpolation resultants.** One-parameter resultants and discriminants are rebuilt from (proved degree bound + 1) exact integer images. *Effect:* captured symbolic A446 pair (degrees 23 and 17, 588/387 terms): 5.888 s with 872 images; every Ducos variant exceeded 60 s. default-on
- A7 **Assignment degree bound and size-derived admission.** The image count uses the smaller of the closed-form Sylvester bounds and an exact assignment (Hungarian) bound on the Sylvester matrix. Plans outside the historical envelope (closed-form bound ≤ 1,024), up to 65,536, are admitted only if an a-priori bit bound on every image fits the caps, so no resource stop can fire after admission. *Effect:* degree-1,828 discriminant: 27.80 s (old multivariate-image interpolation 52.31 s, old default path > 660 s); degree-2,220: 58.83 s against 106.79 s. NEW 2026-10-12 default-on
- A8 **Exact integer images with a zero-image guard.** Each image is an exact integer resultant; an exact gcd test first decides whether it is zero, because Symbolica 3.0.1's Lazard–Ducos panics on a vanishing subresultant (see U3). The same guard protects the full-Ducos path. *Effect:* a pair on which the previous release crashed now solves (7 cells) and verifies. NEW 2026-10-12 default-on
- A9 **Binary (Kronecker) segmentation.** Two-parameter resultants and discriminants are packed into one integer polynomial under proved degree and coefficient bounds, computed once and decoded with complete signed checks. *Effect:* authentic A317/A318 resultants of 9,835/18,939 terms in 38.16/20.03 s; forced on a 56-degree-product discriminant 2.17 s against Ducos 7.80 s. Default since 2026-10-12 (`binary_resultants = false` opts out). NEW 2026-10-12 default-on
- A10 **Second-route spot check.** Every interpolated or segmented result is re-evaluated at one extra full-degree point through Symbolica's own substitution and Brown resultant, in the solver and in the verifier. *Effect:* cost within noise (27.86 s against 27.87 s). NEW 2026-10-12 default-on
- A11 **Sum-variable factor preconditioning.** A small polynomial that depends on two variables only through their sum is factored in fewer variables; the inverse substitution and the factor product are checked. *Effect:* a stalled A446 factorization (native > 60 s) took 0.003–0.025 s. default-on
- A12 **Affine factorization preconditioning.** Before factoring, a bounded exact integer translation moves the polynomial to a regular evaluation fiber; factors are mapped back and the signed product rebuilt. *Effect:* 224-term Gluza33 factor 1.63 s; the native call timed out at 30.02 s. opt-in (`affine_factorization`)
- A13 **Support-preserving diagonal scaling.** Tried first inside the affine policy: nonzero scalings reach the same regular fiber without changing the monomial support. *Effect:* a 1,283-term factor on which translation stalled 30 s factors in 0.048 s. opt-in
- A14 **Bounded exact shear.** Third strategy of the affine policy (`x -> x + c·y` with exact inverse). *Effect:* captured C470 obstruction 0.088 s; the original C470 continuation then saved 56,671 more exact operations (still incomplete). opt-in
- A15 **Primitive parameter-linear irreducibility shortcut.** Gauss's lemma proves a primitive polynomial linear in one variable irreducible without calling the factorizer. *Effect:* 4.840 ⟶ 0.120 ms and 1.463 ⟶ 0.079 ms on two core inputs. default-on

2.2 Root isolation and lifting

- L1 **Algebraic cell boundaries re-isolated at every prefix.** A boundary is a root index of a specialized polynomial; roots are isolated exactly at each new rational prefix and coincident roots are separated by gcd and refinement under a work bound. default-on
- L2 **Open-interval Sturm counting.** Signed primitive pseudo-remainder sequences count roots on open or unbounded intervals (Symbolica only counts on `(0, ∞)`). *Effect:* degree-12 A381 guard 3.184 ⟶ 0.602 ms, largest coefficient 2,281 ⟶ 555 bits. default-on
- L3 **Integer sign evaluation.** Signs at rational points use homogeneous integer Horner evaluation instead of rational arithmetic. *Effect:* 2.64–11.01× in a microbenchmark. default-on
- L4 **Integer prefix evaluation.** Assigned prefixes are substituted through exact integer numerators. *Effect:* guard batches 3.79× (A446) and 5.33× (A381); the matched whole-solver control moved < 1% (43.68 ⟶ 43.30 s). default-on
- L5 **Single-pass batched specialization.** All coordinates of a prefix are substituted in one pass with shared powers. *Effect:* 2,000 prefixes of `ex_e5/2`: 0.401 ⟶ 0.252 s; `ex38`: 0.079 ⟶ 0.052 s. default-on
- L6 **Deferred GCDs and borrowed root signs.** Redundant GCDs are postponed and sign evaluations reuse root data. *Effect:* improves the three cases where symGCAD was slowest relative to GCAD; their time ratios remain 2.09, 1.61 and 2.75. default-on
- L7 **Primitive shared root fibers.** Root intervals share one immutable defining polynomial; retained rational fibers are made primitive integer polynomials; affine prefix guards skip power tables. *Effect:* A414 fixed-work traversal 7.748 ⟶ 6.917 s (–10.7%; a first pair was 5.3% slower). default-on
- L8 **Ordered root union.** A priority queue merges the roots of many cuts with exact equality and bisection checks, including coincident roots. No speedup established. default-on
- L9 **Positive and descending root selectors.** A boundary may count roots among the positive roots only (`positive`) or from the largest real root down (`real_descending`), so collisions among negative roots cannot change its meaning. default-on (produced only by certified rules)

2.3 Projection operators and certified reductions

- P1 **Faithful generic projection.** The `baseline` engine reproduces Strzeboński's projection and lifting exactly (the reference for A1). default-on
- P2 **Optimized projection engine.** The same projection with canonical operation reuse, prefix reuse and certified removals (P3). opt-in (`engine = "optimized"`)

- P3 **Certified factor removal.** A generated projection factor is dropped only with an exact certificate that it is strictly nonzero on the domain (nonnegativity is not enough); original constraints and splits stay protected. `input_positivity` extends this to input factors. *Effect:* fixed A446: 1.326 s with it, timeout at 60 s without. opt-in (`positivity`)
- P4 **Operation pruning.** Tries a nonvanishing proof before factoring a whole projection operation; off by default because a failed proof costs time. opt-in (`prune_operations`)
- P5 **Strict-Bézout pair omission in global projection.** A supplied identity `p*L + q*M = k*S` with `k > 0`, where `S = H^2` with `H` certified nonzero or `S` certified positive, proves that `p, q` have no common zero in the box, so `Res(p, q)` is skipped; p, q, their leading coefficients and discriminants stay, face pairs are never skipped, and each omission is replayed by the verifier. *Effect:* Gluza33 piece `R--`, third level: 1,033 ⟶ 661 polynomials and 228,998 ⟶ 112,759 terms (−36%, −51%), reached at 122.4 s instead of 164.2 s. NEW 2026-10-12 opt-in (`projection_strict_bezout_hints`)

2.4 Positivity and domain certificates

- C1 **Exact nonvanishing certificate portfolio.** Coefficient signs, positive coordinate shifts, Pólya/Bernstein tests, univariate Sturm on open and unbounded domains and exact quadratic rules, all bounded; each certificate is serialized and replayed on the original domain. default-on (used when `positivity` is set)
- C2 **Parameterized quadratics.** Proves positivity of quadratics whose discriminant is exactly negative on the domain. default-on
- C3 **Square plus positive residual.** Writes a polynomial as an exact square (monomial or affine, also after a homogeneous scale is fixed) plus a strictly positive residual, with oriented bounds. default-on
- C4 **Weighted-square certificates.** Sums of positive-monomial-weighted binomial squares plus a positive residual. *Effect:* "positive endpoint" case: 20 ⟶ 1 cell, 37.2 ⟶ 9.3 ms solver time. opt-in (`weighted_square_positivity`)
- C5 **Positive-semidefinite quadratic components.** Exact rational LDL elimination of quadratic subforms plus a strictly positive residual; no floating-point eigenvalues. *Effect:* full quadratic constant 8 ⟶ 1 cell (18.3 ⟶ 7.6 ms); local linear split 12 ⟶ 2 cells, interior zero sheet kept. opt-in (`quadratic_component_positivity`)
- C6 **Caller-chosen certificate budgets.** Library callers may raise the limits to 262,144 Bernstein coefficients and 10^9 work units (defaults 4,096 and 10^6); the chosen limits are serialized and replayed (symGCAD PR #2). opt-in
- C7 **Early abandonment from exact samples.** An exact zero or a sign change at a sample ends a hopeless certificate attempt early; samples never prove anything. default-on
- C8 **Certified local guards.** In the local engine, a certificate replaces a generated guard, and its own premises become protected ancestor obligations. *Effect:* fixed A446 local: 8 certificates replaced 436 guard occurrences; 35 cells in 0.427 s, where the unreduced run timed out at 60 s. opt-in (`positivity` with `engine = "local"`)

2.5 Composite engines

- E1 **Homogeneous positive-scale reduction.** For a homogeneous block, one proved-positive anchor coordinate is fixed to 1; the smaller problem is solved and the original cells are rebuilt exactly. *Effect:* fixed A446 2.409 ⟶ 1.326 s, fixed Ax12 1.032 ⟶ 1.445 s (mixed). opt-in (`homogeneous_block`, `homogeneous_anchor`)
- E2 **Monotone engine.** Peels positive axes on which exact linear or quadratic certificates prove strict increase, solves the endpoint problem and restores every root sector. *Effect:* fixed A446/Ax12: 35 ⟶ 8 cells; A446 1.326 ⟶ 0.825 s. opt-in (`engine = "monotone"`)
- E3 **Conditional quadratic peel.** Positive-multiplier certificates justify consecutive quadratic endpoint reductions. *Effect:* fixed A381 and Ax381: 836 verified cells each, medians 0.186 and 0.102 s; all 6 trials without the rule timed out at 60 s. opt-in (inside `monotone`)
- E4 **Negative-target mode.** Solves `Q > 0` as the negative side of `F = -Q` through its endpoint inequality, restoring only the interval below the positive root. *Effect:* full original physical elliptic problem: 269,061 cells, solve 33.86 s, verify 73.09 s. opt-in (`monotone_negative_target`)
- E5 **Sign filter.** For an unsplit problem, decomposes the full sign arrangement of one non-affine constraint and keeps exactly the required cells, with an exhaustive selection map. opt-in (inside `monotone`)
- E6 **Linear-negative engine.** A target linear in `z` with `A*z + B < 0` is reduced to `A < 0` using a guide whose positivity and square identity `B*C - A*D = k*H^2` are proved; the lower boundary is the largest root of `z*q`. *Effect:* negative Gluza30: 1 cell, 0.185 s; negative Gluza33: 3 cells, 0.164 s solve, 0.021 s verify; GCAD timed out at 60 s on both. opt-in (`engine = "linear_negative"`)
- E7 **Bilinear-positive engine.** For `a*x*y + b*x + c*y + d > 0` on the last two positive axes, seven strict guide certificates fix the fiber shape; the base is solved over the signs of `a, b, c`. *Effect:* positive Gluza30: 80,980 verified cells, solve/verify medians 22.57/16.85 s; GCAD timed out. (Under the opt-in contract X1 the same problem needs 1 cell.) opt-in (`engine = "bilinear_positive"`)
- E8 **General bilinear engine with pieces.** Handles all 16 sign patterns of `a, b, c, d`: ratio premises found by square-free decomposition, a separator `G` with `b*c - a*d = ±m*G^2`, and a planner that splits the base into independently solvable pieces plus proved-empty regions. The verifier checks the partition by a truth table and verifies every piece. *Effect:* Gluza33: plan of 4 pieces + 1 empty claim in 0.034 s; synthetic chain 47 verified cells; under the strict contract the Gluza33 pieces still time out (1,200 s), with X1 the engine completes positive Gluza33. NEW 2026-10-12 opt-in (`engine = "bilinear_general"`)
- E9 **Quadratic ladder.** When a quadratic opens two positive roots below a linear fiber, restores a proved five-sector arrangement over a verified base, keeping both sides of `F = 0`. *Effect:* crossed A446 chamber `t < 4`: 5 verified cells in 0.022 s (debug); the unrestricted problem stays unresolved. opt-in (`engine = "quadratic_ladder"`)
- E10 **Selectable base engines.** Monotone and bilinear wrappers solve their base with `baseline`, `optimized` or `local`, optionally with an inner homogeneous anchor and block that keeps kinematic parameters outside the scale block. opt-in (`bilinear_general_base_block` is NEW 2026-10-12)
- E11 **Projection-free peel inspection.** Reports which peels are certifiable for an input without solving. diagnostic

2.6 Local projection engine

- LP1 **Local projection.** An adaptation of Strzeboński's local projection: cuts are projected only for the stacks actually sampled, missing ancestor cuts are learned and the affected stacks rebuilt; the whole tree is checked by the independent verifier. opt-in (`engine = "local"`)
- LP2 **Batched sibling discovery.** Collects obligations from several siblings before rebuilding a stack. default-on
- LP3 **Direct first-axis base.** On the first lifting axis, univariate cuts need no pair resultants: their real-root union is isolated directly. *Effect:* a fixed A381 run avoided 4,150,168 potential first-axis pairs. default-on
- LP4 **Positive-ray Descartes rule.** On a proved-positive axis, zero or one coefficient sign variation replaces the discriminant (no or exactly one simple positive root). default-on (`local_descartes_one`)
- LP5 **Quadratic positive-ray classifier with cached guards.** Exact quadratic sign cases decide positive roots; guard results are cached. *Effect:* fixed-work A381 debug controls: solver time −15.0% and −14.5% (process wall time mixed). default-on
- LP6 **Leading-term cancellation separation.** Up to two exact cancellations whose coefficients all have one certified sign prove that two polynomials share no positive root, so their resultant is skipped. default-on (`local_pair_cancellations = 2`)
- LP7 **Adjacent-root pairs.** Projects only pairs whose roots are adjacent at the sample. *Effect:* no completion gain: all 27 symbolic release attempts timed out. opt-in (`local_adjacent_pairs`)
- LP8 **Descartes witnesses for 2–8 positive roots.** Rational separators with alternating signs plus coefficient guards prove each interval holds one simple root; at most two templates per polynomial keep the guard set finite; source size budget 1–65,536 terms. *Effect:* with 8,192 terms, the original Gluza33 passed a 6,938-term degree-16 discriminant and saved 4,554 more operations. opt-in (`local_descartes_witnesses`)
- LP9 **Interval Descartes covers.** Covers the positive ray by rational intervals with zero or one sign variation each, for fibers where the global count is not exact. *Effect:* Gluza33 19,942-term degree-24 discriminant (2 roots, 4 variations) bypassed with 49 guards (factored in 9.27 s): +6,422 operations; a 131,072-term aggregate budget gave +7,194; template limit 4 gave +1,094 and +2,397. opt-in (`local_descartes_intervals` and its budgets)
- LP10 **Adjacent interval coalescing.** Merges neighbouring pieces when an exact recount keeps at most one variation. *Effect:* four-root cover with 97 guards / 109,726 terms built in 1.60 s; the greedy cover needed 121 / 139,654, over budget. default-on (when intervals are enabled)
- LP11 **Positive pair witnesses.** Once both polynomials' positive roots are proved delineable, rational separators between them replace the pair resultant. opt-in (`local_positive_pair_witnesses`)

- LP12 **Pair-witness admission gate and axis budget.** A new separator template is admitted only when the resultant it replaces would be large (dense term bound above a threshold), and per-axis template counts can be capped. *Effect:* Gluza33 base piece `a > 0` with witnesses: 20.4 million nodes in 570 s without finishing → 0.02 s, 178 verified cells at gate 4,096; on 19 of 20 corpus cases the gate behaved like witnesses off. Default 0 keeps the old policy. NEW 2026-10-12 opt-in (`local_pair_witness_min_resultant_terms`, `local_pair_witness_axis_budget`)
- LP13 **Local strict-Bézout hints.** The local counterpart of P5 for chosen cut pairs. *Effect:* positive Gluza30: 49 hints skipped 2,918 resultants (no completion). opt-in (`local_strict_bezout_hints`)
- LP14 **Completed-subtree reuse.** Complete conditional proofs are reused at identical rational prefixes after their obligations are rechecked; budget 32 MiB, entries 128–4,096 nodes. *Effect:* C470 at an equal refinement limit 8.40 → 6.96 s with 1,024-node entries; no full-solve gain. default-on
- LP15 **Learned-cut hints.** Unfinished stacks keep their cuts as hints (8 MiB), always re-projected. default-on
- LP16 **Persistent hint archive.** Saves cut hints across restarts (8 MiB). *Effect:* no completion gain in matched restarts. opt-in (`local_hint_archive`)
- LP17 **Stable samples.** Keeps a rational sample across refinements when an exact comparison places it strictly inside a new sector, preserving cache hits. opt-in (`local_stable_samples`)
- LP18 **Deferred projection with cut compaction.** Discovers complete children before projecting, then drops learned cuts no child needs. *Effect:* A414 pilot still timed out at 300 s. opt-in (`local_deferred_projection`)
- LP19 **Retained sibling proofs.** Completed children survive a stack's refinement rounds and move to the new sector containing their unchanged sample, under a shared memory budget (a quarter of the memory limit at 4 KiB per node). *Effect:* Gluza33 piece `Ppm`, 30 min: 12.80 → 1.38 million speculative nodes for the same progress (still incomplete); fixed `B6_469` (witness recipe, gate 4,096) completes only with retention: 112,124 verified cells in 9.91 s. NEW 2026-10-12 opt-in (`local_retain_siblings`)
- LP20 **First-axis square-free support basis.** Replaces irreducible factorization of bounded generated first-axis guards by checked square-free blocks. *Effect:* 92.06 s against 185.83 s on one A377 block set. opt-in (`local_base_univariate_basis`)
- LP21 **Discovery caps.** `local_max_refinements` (10,000) and `local_max_nodes` (100,000) bound speculative work; exhaustion returns `incomplete`. default-on
- LP22 **Refinement-thrash diagnostics.** Read-only counters of restarts, cut origins, discarded children and cache-miss causes (`SYMGCAD_LOCAL_THRASH_DIAG=1`). *Effect:* found 337,116 completed children discarded in 60 s on one Gluza33 piece, which motivated LP12 and LP19. NEW 2026-10-12 diagnostic
- LP23 **Obstruction observer.** Records which guard fails at an exact prefix without changing proofs. diagnostic

2.7 Ordering and search

- O1 **Extra ordering heuristics.** Brown and sparsity orders join GCAD's MODS, SOTD and T1; variable groups and explicit orders bind every candidate. opt-in
- O2 **Bounded MCTS order search.** UCT over legal elimination prefixes (seed 0, at most 256 rollouts, search time `min(300 s, 20%)`, 1/10/60 s promotion tiers); only a completed exact solve can win. *Effect:* 126/126 cold ablation solves verified, but no end-to-end gain over cheap heuristics. opt-in (`ordering = "mcts"`)
- O3 **Search pruning.** Abandons trials whose completed prefixes show severe term growth; a prediction, never a region rejection. default-on (with MCTS)
- O4 **Local-progress promotion score.** Ranks timed-out local trials by polynomial inventory and refinement growth. *Effect:* reduced positive Gluza30 base: 3/3 verified, median 65.10 s, against 3/3 incomplete at 313.63 s. opt-in (`search_local_progress`)
- O5 **Kinematic portfolio ordering.** For parametric Feynman inputs, probes every certified homogeneous anchor × kinematic order in killable workers, counts the exact open kinematic cells, and solves with the fewest; decoupled kinematic variables are lifted last. *Effect:* 26 solved cases: 105.88 → 92.97 s total; C531 21.22 → 2.27 s; 4 held-out cases where the default ran out of 5 GiB now solve (5.7–47.6 s); losses `Cx12_438` 32.30 → 59.82 s and 36 tiny cases 2.07 → 7.73 s. NEW 2026-10-12 opt-in (`ordering = "kinematic_portfolio"`)
- O6 **Reproducible portfolio budgets.** Fixed per-candidate budgets give the same choice on 21/21 cases, at higher cost (1,317 s on the 26 cases). NEW 2026-10-12 opt-in (`portfolio_fixed_budgets`)
- O7 **Order proposal helpers.** Support-order seeds and short linear-prefix probes; measured, no gain. diagnostic

2.8 Caching and checkpointing

- K1 **Exact operation cache.** Factorizations, resultants, discriminants and completed projection prefixes are shared across stages and ordering trials (128 MiB). default-on (optimized and local engines)
- K2 **Persistent checkpoints.** Atomic, hash-validated records of every completed exact operation, kept through interruptions mid-stage. *Effect:* the final original Gluza33 campaign retained 32,456 operations (4,559 new) for continuation. opt-in (`checkpoint_dir`; automatic with MCTS)
- K3 **Domain-free operation keys.** Pure algebra records are keyed by variable map, axis, backend version and exact input, no longer by the domain; old records are validated and migrated. *Effect:* Gluza33 `R--` after `R+-`: 50 of 127 operations from disk through level 2 (old scheme: 0). NEW 2026-10-12 default-on (with `checkpoint_dir`)
- K4 **Primitive-linear cache bypass.** Trivially irreducible factors skip cache I/O. *Effect:* 30 polynomials, fresh memory and disk caches: 15.531 → 0.415 ms. default-on
- K5 **Search journal.** MCTS state (8 MiB, 4,096 records) is replayed for the same problem and executable; replayed claims count as unverified until re-solved. default-on (with MCTS)
- K6 **Untrusted proposal caches.** First-axis support proposals and hint archives are re-factored and re-checked on load; verification never reads a cache. opt-in
- K7 **Cache integrity audits.** Inherited bytes, operation keys and output hashes are checked before a continuation reuses work. diagnostic

2.9 Verification and evidence

- V1 **Explicit generic-coverage contract.** A result proves that its cells lie in the solution set `S` and cover `S` up to the zero set of an explicit finite list of nonzero exceptional polynomials; GCAD returns cells without this list. default-on
- V2 **Self-contained JSON proof.** Records the normalized problem, actual order, polynomial table, projection levels, rational samples, root-index boundaries, an exhaustive tree of stacks, rejections and accepted cells, and one of four statuses (`complete_generic`, `incomplete`, `invalid_input`, `computation_failure`). default-on
- V3 **Independent verifier.** `symgcad verify` re-normalizes the input, rebuilds every projection and reduction obligation, checks roots with exact arithmetic and walks every open interval; every engine has its own typed replay; an incomplete result cannot pass. default-on
- V4 **Raw cells and seams by default.** No merging unless asked: interior `F = 0` sheets and same-sign seams always remain boundaries (GCAD merges aggressively). default-on
- V5 **Indexed compatibility merge.** GCAD-style merging as a separate output that records the excluded seams. *Effect:* the 131,256-cell ex38 now completes (median 40.72 s, 4.77 s merging; GCAD 26.50/7.45 s). opt-in (`merge = "aggressive"`)
- V6 **Replay hardening for the new rules.** Strict-Bézout omissions (11 tampered variants rejected), `bilinear_general` evidence (about 40 forgeries rejected), the assignment degree bound (checked through its LP dual) and unresolved portfolio inputs are all checked or refused by the verifier. NEW 2026-10-12 default-on
- V7 **Certified comparison of decompositions.** Exact common refinement of two results, across different orders and reductions. diagnostic
- V8 **Seeded interior sanity check.** Fresh rational points in every cell, checked against the original TOML expressions. *Effect:* fixed A446: 8 cells, 16 `F` and 112 constraint evaluations, 0 failures, 0.41 s. diagnostic
- V9 **Native comparison harness and parity audit.** Frozen GCAD cohort, three paired runs, adaptive native cap `max(300 s, 3 × symGCAD solve)`; documents GCAD merge and assertion defects (e.g. `failing_test_ex43`). diagnostic
- V10 **Signed grid verification of huge algebra.** The A377 degree-32 discriminant candidate (degree 4,058, 3,261 terms) was proved by 4,773 exact evaluation points in 2,093.9 s. diagnostic
- V11 **Original-identity audit.** Every claimed completion is re-checked against the original polynomials, domain, groups and order: 303 of 331 distinct problems (124/124 fixed ZZ, 97/124 symbolic ZZ, 82/83 core). diagnostic

2.10 Robustness and supervision

- R1 **Killable worker processes.** Solves, verifications and probes run in separate process groups with hard wall-time and memory limits (defaults 1 h, 64 GiB, 16 workers; 2 h on request), so a blocking native call cannot hang the run. default-on
- R2 **Campaign and slow-suite runners.** `symgcad bench` and the resumable one-hour slow queue record wall, CPU and RSS; timeouts stay incomplete. default-on
- R3 **Fast regression suite.** 753 tests at `9baf91c` (744 at `4dbe440`, 662 at `81d9ac5`), including proof mutations, same-sign seams, degree drops and worker termination. default-on
- R4 **Operand capture and replay.** `SYMGCAD_CAPTURE_DIR` saves the exact operand of the latest uncached operation (16 MiB cap) for offline profiling. diagnostic
- R5 **Throttled progress reporting.** *Effect:* an unthrottled report had cost 17.505 s of system CPU in a 60 s C470 trial; now 1.282 s. default-on
- R6 **Buffered result export.** *Effect:* an A286 run that failed through 392,791 tiny writes completes 3,308 cells in 1.354 s. default-on
- R7 **Interrupted-worker accounting.** Timed-out workers keep their elapsed time and typed diagnostics. default-on
- R8 **Bounded parsing of untrusted hint text.** Size, degree and coefficient bounds are checked on the parse tree before Symbolica expands anything. *Effect:* `(x+y+z+1)^300` was rejected after 6.9 s, now in < 0.5 s; `7^(3^20)`, which never returned, is rejected. NEW 2026-10-12 default-on
- R9 **Fully supervised portfolio and pieces.** Portfolio enumeration, probes and `pieces plan/assemble` run in bounded workers; any failure falls back to MODS with the reason recorded. *Effect:* 3 s-wall test stops at 3.08 s with 491–518 MB peak (before: 19.1 s, 1.97 GB). NEW 2026-10-12 default-on
- R10 **Reproducible environment.** Nix flake and Cargo lock with a pinned public Symbolica revision; no private checkout needed (symGCAD PR #1); CPU-affinity tools for pinned measurements. default-on

2.11 Interfaces

- I1 **Rust library and TOML command line.** `solve`, `verify`, `project`, `order`, `bench`, `corpus`, `import-fastsecdec`. Inputs take exact rationals, `[parameters]`, `variable_groups`, binding orders and `[limits]`; unknown fields are refused (GCAD: a Python API on SymPy expressions). default-on
- I2 **`project` matches `solve`.** `symgcad project` now exports exactly the projection that `solve` lifts, including homogeneous reduction. *Effect:* Gluza33 `R+-`: 354.5 s / 453 MB, before a timeout at 1,500 s. NEW 2026-10-12 default-on
- I3 **Piece-wise solving from the command line.** `symgcad pieces plan` writes one TOML per piece; pieces are solved as separate processes in any order; `pieces assemble` checks and joins them. NEW 2026-10-12 opt-in
- I4 **FastSecDec graph import.** Reads native signed `U/F`, exact kinematics and edge-to-parameter maps from dressed HEPKit DOT graphs and checks them before solving. opt-in
- I5 **Rust corpus builder.** Turns GCAD's 177 collected tests, 265 literal two-loop and 12 generated one-loop fixtures into 331 exactly deduplicated problems (124 fixed ZZ, 124 symbolic ZZ, 83 core). default-on
- I6 **Generated benchmark document.** This report: `benchmark/build_report.py` turns the cold symGCAD runs, the native GCAD runs, the long attempts and the FLINT measurements into `report-data.json` with per-row provenance; Typst renders it. `benchmark/run_final.sh` repeats the whole campaign with one command. NEW 2026-10-12 default-on
- I7 **Reproducible full-corpus campaign.** One manifest of the 331 groups and their recipes, a cold symGCAD runner (fresh checkpoint directory and `TMPDIR` per run, one pinned CPU, independent `verify`) and a native runner (unmodified GCAD through the unmodified bridge, 300 s per sign case, 8 GiB). Both are resumable and record binary hashes. NEW 2026-10-12 diagnostic
- I8 **Symbolica vs FLINT harness.** Exact operands captured from real solves are replayed through Symbolica's production kernels and FLINT 3.6 with canonical output comparison (Section 3). NEW 2026-10-12 diagnostic

2.12 Upstream dependency fixes

- U1 **Symbolica PR #69: resultant sign with a linear argument.** `Res(x^3+y, x+2)` returned `y-8` instead of `8-y` in every public strategy. symGCAD uses its own signed identity for linear operands; the patch is optional. optional patch
- U2 **Symbolica PR #75: integer left-shift overflow.** `Integer::from(1) << 63` gave a negative number; the patch is checked by 546 comparisons. No production path was affected. NEW 2026-10-12 (PR opened) optional patch
- U3 **Symbolica PR #76: Lazard–Ducos panic on a vanishing subresultant.** Explicit zero coefficients made Ducos index out of bounds; patch with a regression test that fails before and passes after; production avoids it through A8. NEW 2026-10-12 optional patch
- U4 **Further upstream reports.** PRs #61 (determinant sign on odd row swaps), #67 and #68 (F4 Gröbner zero generators and cache), #73 (zero-variable exponent iteration), found while auditing Symbolica; no local patch is needed. upstream only
- U5 **Bounded small-leading-coefficient Wang factorization.** *Effect:* 2,410-term Gluza33 factor 1.039 s, against a 60.03 s timeout without it. optional patch (experimental)
- U6 **Bounded main-axis choice for factorization.** *Effect:* 1,731-term component 0.280 s against a 30.03 s timeout; the whole 2,867-term source 1.951 s. optional patch (experimental)
- U7 **Positive-only isolation and unit Taylor shift.** *Effect:* unit shift −39% on an authentic operand; positive-root-only isolation 557.98 s / 421 MiB against 765.81 s / 813 MiB for all roots. optional patch (research backend)
- U8 **Idempotent patch bootstrap.** `scripts/bootstrap-deps.sh --apply-patches` applies the optional fixes to a separate research checkout; the default build stays pristine. opt-in

2.13 Opt-in contract extensions

- X1 **Projective sections (opt-in contract extension).** With `solver.projective_sections = true` (engine `bilinear_general` only, default off), a cell's upper boundary may be "the k-th positive root of P, or +∞ where P has fewer roots": inside one cell the boundary can pass continuously through infinity (cells stay connected and cylindrical after the map `u = x/(1+x)`). For `F = a*x*y + b*x + c*y + d` with the replayed ratio premises and `b*c - a*d = m*G^2`, the fiber over a base point is empty when `d < 0` and exactly `{0 < y < Y*, 0 < x < X(y)}` when `d > 0`, so the base only has to decompose `d > 0`, one cell per base cell. The verifier replays the premises and the plan; every other verifier, the merge and the cell comparison refuse or handle such bounds explicitly. Approved by the user on 2026-10-12. *Effect:* positive Gluza33 (unresolved under the strict contract, longest attempt 7,200 s): 2,372 verified cells, solve 0.42 s, verify 0.55 s; a seeded sanity check of 4,744 fresh points evaluated the original `F` exactly with 0 failures. Positive Gluza30: 1 cell instead of 80,980. Not counted in the strict totals. NEW 2026-10-12 opt-in
- X2 **Aggressive merging.** GCAD-compatible merging that may cross seams; only as a separate output (see V5). opt-in

2.14 Investigated, not adopted

- N1 **Convex two-point theory (2026-10-12).** For two-point graphs, `F/U` is convex; dropping touching pairs shrinks the Gluza33 level-2 projection from 86 to 71 factors under the current cell contract, and to 34 only with "generalized sectors", a contract change that was not implemented. diagnostic
- N2 **Relevance-restricted projection prototype (2026-10-12).** No-go: about 1.9 million base and 3.0 million 8-variable cells for positive Gluza33 under the strict contract (ordinary CAD: about 10^{10} on one piece). diagnostic
- N3 **Literature and Symbolica audits (2026-10-12).** Landau, PLD and Whitney sets are necessary but not sufficient for cylindrical cells; Symbolica already supplies every primitive used, and over `Q(√10)` it splits Gluza33 discriminant factors (450 ⟶ 43 × 43 terms in 1.05 s) but stalls on 2,250 terms. diagnostic
- N4 **Domain-aware ordering profile (2026-10-12).** Anchors and orders picked by predicted kinematic cells made solved symbolic test beds 2–312× faster (led to O5); no effect on the unresolved two-loop cases. diagnostic
- N5 **Earlier research lines.** CFF and Landau chambers, tropical and SubTropica ordering, ordered quadratic flag identities, crossing transport, deterministic CRT and factorial-convolution Taylor shifts produced exact identities or proposals but no production gain. diagnostic

3 Symbolica vs FLINT on the algebraic steps

symGCAD runs all exact polynomial algebra through Symbolica; the original GCAD calls FLINT. To isolate the algebra from everything else, exact operands were captured from real symGCAD solves (Gluza30/33, A317, A5_31, fixed A377 and others), written in a neutral format, and replayed one operation per fresh process through (i) the kernel symGCAD's production code actually uses (Symbolica, after symGCAD's own dispatch: closed-form discriminants, interpolation, binary segmentation), (ii) Symbolica's individual algorithms (Lazard–Ducos subresultants, Brown/CRT), and (iii) FLINT 3.6 (`fmpz_mpoly` and `fmpz_poly`). Every output pair is compared after canonicalization (signs and content); 164 outputs were compared with 0 mismatches. Times are the minimum over repetitions on one pinned CPU; **ratio = Symbolica time / FLINT time**, so a ratio above 1 means FLINT is faster.

RESULTANT
1.02
geometric-mean ratio over 17 of 67 operands; total-time ratio 1.12

DISCRIMINANT
0.83
geometric-mean ratio over 12 of 34 operands; total-time ratio 0.13

FACTOR
2.38
geometric-mean ratio over 18 of 42 operands; total-time ratio 11.64

GCD
3.77
geometric-mean ratio over 4 of 24 operands; total-time ratio 18.32

Reading. By geometric mean: resultant about even (1.02); discriminant about even (0.83); factor FLINT faster (2.38); gcd FLINT faster (3.77). The geometric mean weights every operand equally, so it is dominated by small, sub-millisecond operations; the total-time ratio is dominated by the few largest operands. Resultants and discriminants matter most for projection; factorization and GCDs build its square-free bases. Operands without a ratio did not finish on both sides within the limits of the snapshot, or were still queued; they are listed in the tables below. Best case for Symbolica: `g33s2-dis-20-569a4bfe48` (ratio 0.13); best case for FLINT: `g33s2-fac-28-3a75cafd52-gcd` (ratio 18.4). The per-operand data, the harness and the reproduction commands are in `benchmark/FLINT_vs_symbolica/`. This section shows a draft snapshot (`benchmark/flint-draft/`) of a campaign that was still running; the report switches automatically to `benchmark/FLINT_vs_symbolica/results/` once that directory exists.

Operation	n	with both times	geo. mean ratio	total-time ratio
resultant	67	17	1.02	1.12
discriminant	34	12	0.83	0.13
factor	42	18	2.38	11.6
gcd	24	4	3.77	18.3
Z[x] resultant (interp. image)	2	0	–	–
Z[x] resultant (segmentation)	26	0	–	–

Table 1: Symbolica (symGCAD production path) vs FLINT 3.6.0 by operation type.

Operation	Size class	n	Production / FLINT (geo. mean)	Range	Symbolica faster	Ducos / FLINT Ducos
resultant	<1 ms	17	1.02	0.54–2.74	10/17	1.02 (17)
resultant	timeout	50	–	–	0/0	–
discriminant	<1 ms	10	0.96	0.55–1.71	6/10	1.13 (9)
discriminant	1-100 ms	1	1.24	1.24–1.24	0/1	1.23 (1)
discriminant	>=10 s	1	0.13	0.13–0.13	1/1	0.74 (1)
discriminant	timeout	22	–	–	0/0	–
factor	<1 ms	9	0.98	0.27–2.17	5/9	–
factor	1-100 ms	8	5.27	1.16–14.2	0/8	–
factor	>=10 s	1	11.7	11.7–11.7	0/1	–
factor	timeout	24	–	–	0/0	–
gcd	<1 ms	2	2.14	2.09–2.19	0/2	–
gcd	1-100 ms	1	2.39	2.39–2.39	0/1	–
gcd	0.1-10 s	1	18.4	18.4–18.4	0/1	–
gcd	timeout	20	–	–	0/0	–
Z[x] resultant (interp. image)	timeout	2	–	–	0/0	–
Z[x] resultant (segmentation)	timeout	26	–	–	0/0	–

Table 2: Symbolica / FLINT time ratios by operation type and size class (class from the slower of the two).

Operand	Active vars; degrees	Terms	Production kernel	Symbolica production	FLINT	Ratio	Symbolica Ducos
g33-disc2-interpimg	X; (36); (35)	37+36	–	–	–	–	–
g33-disc3-interpimg	X; (42); (41)	43+42	–	–	–	–	–
a317-pair-segenc	X; (12); (12)	13+13	–	–	–	–	–
a318-pair-segenc	X; (10); (6)	11+7	–	–	–	–	–

Operand	Active vars; degrees	Terms	Production kernel	Symbolica production	FLINT	Ratio	Symbolica Ducos
a377-1d3e6e9835cc-segenc	X; (12); (8)	13+9	–	–	–	–	–
a377-1d4048e35d3e-segenc	X; (8); (8)	9+9	–	–	–	–	–
a377-2084c3ad850e-segenc	X; (8); (12)	9+13	–	–	–	–	–
a377-22488f0f86d7-segenc	X; (8); (8)	9+9	–	–	–	–	–
a377-27020a42fd8e-segenc	X; (8); (8)	9+9	–	–	–	–	–
a377-4c165beeeaaa-segenc	X; (8); (8)	9+9	–	–	–	–	–
a377-87adaef31346-segenc	X; (8); (8)	9+9	–	–	–	–	–
a377-9678884cb7f0-segenc	X; (8); (11)	9+12	–	–	–	–	–
a377-ac6f3048ce62-segenc	X; (12); (12)	13+13	–	–	–	–	–
a377-cc41b49c69bd-segenc	X; (12); (8)	13+9	–	–	–	–	–
g33S2-dis-20-569a4bfe48-segenc	X; (12); (11)	13+12	–	–	–	–	–
g33S2-dis-21-d3fad47e59-segenc	X; (12); (11)	13+12	–	–	–	–	–
g33S2-dis-23-8acb497b1c-segenc	X; (10); (9)	11+10	–	–	–	–	–
g33S2-dis-24-398386d3e8-segenc	X; (10); (9)	11+10	–	–	–	–	–
g33S2-res-00-11a890ffaa-segenc	X; (12); (12)	13+13	–	–	–	–	–
g33S2-res-01-92460725ec-segenc	X; (10); (12)	11+13	–	–	–	–	–
g33S2-res-02-41f64e7684-segenc	X; (8); (12)	9+13	–	–	–	–	–
g33S2-res-03-f3f5c3be5d-segenc	X; (12); (10)	13+11	–	–	–	–	–
g33S2-res-04-1b1af7b175-segenc	X; (12); (10)	13+11	–	–	–	–	–
g33S2-res-05-83bfdd39e0-segenc	X; (10); (12)	11+13	–	–	–	–	–
g33S2-res-06-e4dacfd3fb-segenc	X; (12); (10)	13+11	–	–	–	–	–
g33S2-res-07-1c09fee64d-segenc	X; (12); (8)	13+9	–	–	–	–	–
g33S2-res-08-17eaa2af35-segenc	X; (12); (10)	13+11	–	–	–	–	–
g33S2-res-09-6ab3116dc1-segenc	X; (10); (10)	11+11	–	–	–	–	–
g33-disc1	x7,x2,x1; (10,10,8)	474	–	–	–	–	–
g33-disc2	x7,x2; (44,36)	999	–	–	–	–	–
g33-disc3	x7,x2; (47,42)	1190	–	–	–	–	–
g33-disc4	x7,x2; (88,68)	3795	–	–	–	–	–
g33S2-dis-20-569a4bfe48	x1,x2,x7; (12,12,12)	993	binary-segmentation	37.2 s	278 s	0.13	206 s
g33S2-dis-21-d3fad47e59	x1,x2,x7; (8,12,12)	679	–	–	–	–	–
g33S2-dis-22-60a726bccc	x1,x2,x7; (12,8,12)	679	–	–	–	–	–
g33S2-dis-23-8acb497b1c	x1,x2,x7; (10,10,12)	705	–	–	–	–	–
g33S2-dis-24-398386d3e8	x1,x2,x7; (10,10,11)	630	–	–	–	–	–
g33S2-dis-25-cc11b6eb47	x1,x2,x7; (4,6,6)	116	–	–	–	–	–
g33S2-dis-26-fbcef3db6c	x1,x2,x7; (3,4,4)	43	–	–	–	–	–
g33S2-dis-27-60776320bd	x0,x1,x2,x6,x7; (2,2,2,2,2)	36	–	–	–	–	–
g33S2-inprogress-dis-08eb8141fe	x1,x7; (100,128)	7969	–	–	–	–	–
symA317-dis-10-d9e5bd74c8	s,t,x0,x2; (6,4,4,4)	229	–	–	–	–	–
symA317-dis-11-17b4714dbc	s,t,x0,x2; (4,2,4,4)	126	–	–	–	–	–
symA317-dis-12-f4fc65cbe2	s,t,x0,x2,x3; (2,1,3,3,4)	61	–	–	–	–	–
symA317-dis-13-8b57471a76	s,t,x0,x2; (3,2,2,2)	42	–	–	–	–	–
symA317-dis-14-5981f9d905	s,x0,x2,x3; (1,2,2,2)	12	–	–	–	–	–
symA317-inprogress-dis-55c0861a04	s,t,x0,x2; (9,6,8,8)	1307	–	–	–	–	–
symA5_31-dis-06-3281fbd1c8	s,t,x3; (3,2,2)	13	–	–	–	–	–
symA5_31-dis-07-c18ffcbe67	s,t,x2,x3; (2,1,2,2)	13	–	–	–	–	–
symA5_31-dis-08-0e8c74f737	s,t,x0,x1,x2,x3; (1,1,2,2,2,2)	17	–	–	–	–	–
symA5_31-dis-09-fc30170f80	s,t; (2,2)	8	–	–	–	–	–
g33S2-fac-28-3a75cafd52	x1,x7; (228,202)	21469	symgcad-factor-basis	10.6 s	905.8 ms	11.7	–
g33S2-fac-29-741a8fb652	x1,x7; (208,186)	18643	–	–	–	–	–
g33S2-fac-30-3c86713b00	x1,x7; (184,168)	14995	–	–	–	–	–
g33S2-fac-31-70f8238aed	x1,x7; (208,180)	16461	–	–	–	–	–
g33S2-fac-32-cc03af2bf8	x1,x7; (208,184)	17169	–	–	–	–	–
g33S2-fac-33-d966d83ef5	x1,x7; (206,182)	18211	–	–	–	–	–
g33S2-fac-34-5056e9cf6a	x1,x7; (184,172)	14805	–	–	–	–	–
g33S2-fac-35-b4319ee36c	x1,x7; (168,160)	12905	–	–	–	–	–
g33S2-fac-36-b41f46e3ca	x1,x7; (64,64)	2145	–	–	–	–	–
g33S2-fac-37-12e6b014e4	x1,x7; (56,52)	1353	–	–	–	–	–
g33S2-fac-38-fd912ec349	x1,x7; (34,36)	753	–	–	–	–	–

Operand	Active vars; degrees	Terms	Production kernel	Symbolica production	FLINT	Ratio	Symbolica Ducos
g33S2-fac-39-035774c28d	x1,x7; (28,28)	425	–	–	–	–	–
g33S2-fac-40-2e7bf33ca7	x1,x7; (23,21)	253	–	–	–	–	–
g33S2-fac-41-a2d1fd70cf	x1,x7; (10,12)	87	–	–	–	–	–
symA317-fac-15-83da5119a0	s,t,x0,x2; (11,6,12,12)	2553	–	–	–	–	–
symA317-fac-16-37628a37c1	s,t,x0,x2; (12,8,6,6)	942	–	–	–	–	–
symA317-fac-17-4521c9cb95	s,t,x0,x2; (6,2,6,6)	312	–	–	–	–	–
symA317-fac-18-b3e6617037	s,t,x0,x2; (6,4,4,4)	285	–	–	–	–	–
symA317-fac-19-593adb827e	s,t,x0,x2,x3; (2,2,1,1,2)	17	–	–	–	–	–
symA5_31-fac-10-b7e40fd7bb	s,t,x3; (6,4,4)	59	–	–	–	–	–
symA5_31-fac-11-b93dca9a03	s,t,x2,x3; (4,2,4,4)	67	–	–	–	–	–
symA5_31-fac-12-4262afaab8	s; (10)	11	–	–	–	–	–
symA5_31-fac-13-b08fda434b	s,t,x3; (6,4,4)	52	–	–	–	–	–
symA5_31-fac-14-9ed62b27e5	s,x3; (2,2)	5	–	–	–	–	–
symA5_31-fac-15-7db2f4e614	s; (6)	3	–	–	–	–	–
g33S2-fac-29-741a8fb652-gcd	x1,x7; (208,186); (207,186)	18643+18643	–	–	–	–	–
g33S2-fac-30-3c86713b00-gcd	x1,x7; (184,168); (183,168)	14995+14995	–	–	–	–	–
g33S2-fac-31-70f8238aed-gcd	x1,x7; (208,180); (207,180)	16461+16461	–	–	–	–	–
g33S2-fac-32-cc03af2bf8-gcd	x1,x7; (208,184); (207,184)	17169+17169	–	–	–	–	–
g33S2-fac-33-d966d83ef5-gcd	x1,x7; (206,182); (205,182)	18211+18211	–	–	–	–	–
g33S2-fac-34-5056e9cf6a-gcd	x1,x7; (184,172); (183,172)	14805+14805	–	–	–	–	–
g33S2-fac-35-b4319ee36c-gcd	x1,x7; (168,160); (167,160)	12905+12905	–	–	–	–	–
g33S2-fac-36-b41f46e3ca-gcd	x1,x7; (64,64); (63,63)	2145+2080	–	–	–	–	–
g33S2-fac-37-12e6b014e4-gcd	x1,x7; (56,52); (55,52)	1353+1353	–	–	–	–	–
g33S2-fac-38-fd912ec349-gcd	x1,x7; (34,36); (34,35)	753+720	–	–	–	–	–
g33S2-fac-39-035774c28d-gcd	x1,x7; (28,28); (27,28)	425+425	–	–	–	–	–
g33S2-fac-40-2e7bf33ca7-gcd	x1,x7; (23,21); (22,21)	253+253	–	–	–	–	–
g33S2-fac-41-a2d1fd70cf-gcd	x1,x7; (10,12); (10,11)	87+77	–	–	–	–	–
symA317-fac-15-83da5119a0-gcd	s,t,x0,x2; (11,6,12,12); (11,6,11,11)	2553+2401	–	–	–	–	–
symA317-fac-16-37628a37c1-gcd	s,t,x0,x2; (12,8,6,6); (11,8,6,6)	942+942	–	–	–	–	–
symA317-fac-17-4521c9cb95-gcd	s,t,x0,x2; (6,2,6,6); (5,2,6,6)	312+228	–	–	–	–	–
symA317-fac-18-b3e6617037-gcd	s,t,x0,x2; (6,4,4,4); (5,4,4,4)	285+210	–	–	–	–	–
symA5_31-fac-10-b7e40fd7bb-gcd	s,t,x3; (6,4,4); (5,4,4)	59+50	–	–	–	–	–
symA5_31-fac-11-b93dca9a03-gcd	s,t,x2,x3; (4,2,4,4); (3,2,4,4)	67+44	–	–	–	–	–
symA5_31-fac-13-b08fda434b-gcd	s,t,x3; (6,4,4); (5,4,4)	52+47	–	–	–	–	–
a317-pair	s,t,x0; (10,6,12); (10,6,12)	324+323	–	–	–	–	–
a318-pair	s,t,x1; (22,20,10); (8,8,6)	2560+161	–	–	–	–	–
a377-1d3e6e9835cc	s,t,x0; (16,6,12); (13,6,8)	1009+575	–	–	–	–	–
a377-1d4048e35d3e	s,t,x0; (14,7,8); (16,8,8)	783+885	–	–	–	–	–
a377-2084c3ad850e	s,t,x0; (10,4,8); (14,4,12)	322+590	–	–	–	–	–
a377-22488f0f86d7	s,t,x0; (16,8,8); (13,6,8)	885+575	–	–	–	–	–
a377-27020a42fd8e	s,t,x0; (14,7,8); (15,6,8)	783+770	–	–	–	–	–
a377-4c165beeeaaa	s,t,x0; (16,8,8); (15,6,8)	885+770	–	–	–	–	–
a377-7b0a2b721fd6	s,t,x0; (11,6,6); (13,6,8)	403+575	–	–	–	–	–
a377-87adaef31346	s,t,x0; (16,8,8); (18,8,8)	885+915	–	–	–	–	–
a377-9678884cb7f0	s,t,x0; (10,4,8); (12,7,11)	324+719	–	–	–	–	–
a377-ac6f3048ce62	s,t,x0; (16,6,12); (14,4,12)	1009+590	–	–	–	–	–
a377-cc41b49c69bd	s,t,x0; (16,6,12); (8,0,8)	1009+61	–	–	–	–	–
g33-final-stalled	x2,x1,x7; (4,3,4); (42,32,20)	43+16828	–	–	–	–	–
g33S2-res-00-11a890ffaa	x1,x2,x7; (8,12,12); (12,12,12)	679+993	–	–	–	–	–
g33S2-res-01-92460725ec	x1,x2,x7; (10,10,12); (12,12,12)	705+993	–	–	–	–	–
g33S2-res-02-41f64e7684	x1,x2,x7; (12,8,12); (12,12,12)	679+993	–	–	–	–	–
g33S2-res-03-f3f5c3be5d	x1,x2,x7; (12,12,12); (10,10,11)	993+630	–	–	–	–	–
g33S2-res-04-1b1af7b175	x1,x2,x7; (8,12,12); (10,10,12)	679+705	–	–	–	–	–
g33S2-res-05-83bfdd39e0	x1,x2,x7; (8,10,10); (12,12,12)	457+993	–	–	–	–	–
g33S2-res-06-e4dacfd3fb	x1,x2,x7; (8,12,12); (10,10,11)	679+630	–	–	–	–	–
g33S2-res-07-1c09fee64d	x1,x2,x7; (8,12,12); (12,8,12)	679+679	–	–	–	–	–
g33S2-res-08-17eaa2af35	x1,x2,x7; (8,12,12); (8,10,10)	679+457	–	–	–	–	–
g33S2-res-09-6ab3116dc1	x1,x2,x7; (10,10,12); (10,10,11)	705+630	–	–	–	–	–

Operand	Active vars; degrees	Terms	Production kernel	Symbolica production	FLINT	Ratio	Symbolica Ducos
g33S2-res-10-213a6a967e	x1,x2,x7; (10,4,10); (12,12,12)	294+993	–	–	–	–	–
g33S2-res-11-15e3a18833	x1,x2,x7; (4,6,6); (8,8,8)	116+303	–	–	–	–	–
g33S2-res-12-0bcabfb761	x1,x2,x7; (4,3,4); (8,10,10)	47+457	–	–	–	–	–
g33S2-res-13-9b5ef38ded	x1,x2,x7; (4,4,6); (4,6,6)	76+101	–	–	–	–	–
g33S2-res-14-58076074a3	x1,x2,x7; (4,4,4); (4,6,6)	57+101	–	–	–	–	–
g33S2-res-15-d4bb1bb14e	x1,x2,x7; (2,2,2); (8,8,8)	7+303	–	–	–	–	–
g33S2-res-16-ce8f161c86	x1,x2,x7; (2,4,4); (4,4,4)	31+57	–	–	–	–	–
g33S2-res-17-fc148e65de	x1,x2,x7; (2,3,3); (4,3,4)	22+47	–	–	–	–	–
g33S2-res-18-ee2240aac4	x1,x2,x7; (2,2,2); (4,4,4)	13+42	–	–	–	–	–
g33S2-res-19-21abb3fee9	x1,x2,x7; (2,2,2); (2,3,3)	10+25	–	–	–	–	–
symA317-res-00-c0eed4af4e	s,t,x0,x2; (6,4,4,4); (9,6,8,8)	229+1307	–	–	–	–	–
symA317-res-01-6bbe449f39	s,t,x0,x2; (4,2,4,4); (9,6,8,8)	126+1307	–	–	–	–	–
symA317-res-02-40ca33722b	s,t,x0,x2; (4,2,4,4); (6,4,4,4)	126+229	–	–	–	–	–
symA317-res-03-db3a4322d9	s,t,x0,x2; (2,0,2,2); (9,6,8,8)	10+1307	–	–	–	–	–
symA317-res-04-84e372f8b0	s,t,x0,x2; (3,2,2,2); (9,6,8,8)	42+1307	–	–	–	–	–
symA317-res-05-9a9134f563	s,t,x0,x2; (3,2,2,2); (9,6,8,8)	42+1307	–	–	–	–	–
symA317-res-06-ff0b838eb1	s,t,x0,x2; (3,2,2,2); (4,3,3,2)	42+72	–	–	–	–	–
symA317-res-07-c1b2d66c5c	s,t,x0,x2; (4,2,4,4); (1,0,2,2)	126+8	–	–	–	–	–
symA317-res-08-4dd94dfa1e	s,t,x0,x2,x3,x5,x8; (0,0,0,0,0,0,1); (1,1,2,2,2,2,1)	1+42	–	–	–	–	–
symA317-res-09-b15af2e0db	s,x0,x2; (1,2,2); (1,2,2)	8+4	–	–	–	–	–
symA5_31-res-00-dee965d81f	s,t,x1,x2,x3; (2,1,1,2,2); (1,1,2,2,2)	11+11	–	–	–	–	–
symA5_31-res-01-9aba7cdcc3	s,t,x2,x3; (2,1,2,2); (2,0,2,2)	13+7	–	–	–	–	–
symA5_31-res-02-ba5210d195	s,t; (6,2); (3,2)	18+7	–	–	–	–	–
symA5_31-res-03-fc97cfbbef	s,t,x3; (3,2,2); (4,2,2)	13+16	–	–	–	–	–
symA5_31-res-04-8f6ff15344	s,t; (1,2); (4,2)	4+10	–	–	–	–	–
symA5_31-res-05-134dc9324f	s,t; (4,2); (1,1)	12+3	–	–	–	–	–

Table 3: Large and notable operands (timeouts: 600 s production/FLINT, 300 s alternative kernels).